

AI-DRIVEN RECRUITMENT ECOSYSTEM FOR INTELLIGENT JOB
MATCHING AND PREDICTIVE CAREER DEVELOPMENT

RESEARCH COMPONENT 1 FINAL DISSERTATION DRAFT

AUTOMATED SEMANTIC RESUME
SCREENING AND IT JOB ROLE
CLASSIFICATION

An Auditable Multi-Stage Ingestion Pipeline with Boundary-Guarded Named Entity
Recognition, Granular Experience Disambiguation, Two-Dimensional Credential
Decoupling, and Balanced Multinomial Logistic Role Classification

Author / Researcher:	D. T. D. Perera
Student Identification Number:	IT22089236
Academic Degree Program:	B.Sc. (Hons) in Information Technology Specializing in Software Engineering
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A dissertation draft submitted in partial fulfillment of the requirements for the degree of Bachelor of Science (Honours) in
Information Technology at the Sri Lanka Institute of Information Technology.

DECLARATION OF ORIGINALITY

I hereby declare that this dissertation titled '**Automated Semantic Resume Screening and IT Job Role Classification**' is the record of authentic research work carried out by me under research project identifier **R26-IT-148** in partial fulfillment of the requirements for the Bachelor of Science (Honours) in Information Technology degree at the Sri Lanka Institute of Information Technology (SLIIT).

I confirm that the content presented in this report is my original intellectual work and has not been submitted previously in whole or in part for the award of any other degree, diploma, or academic fellowship at this or any other tertiary educational institution.

All algorithms, mathematical models, experimental methodologies, dataset evaluation figures, and software source codes described in this dissertation are authentic. In strict compliance with academic research integrity, no experimental results, F1-scores, accuracy figures, or citations have been fabricated or artificially altered. Where information and secondary models have been derived from published academic literature, they have been comprehensively cited in the references section.

Candidate Name: D. T. D. Perera

Candidate Student ID: IT22089236

Signature of Candidate:

Date: October 10, 2026

SUPERVISOR ENDORSEMENT

This is to certify that this research dissertation titled '**Automated Semantic Resume Screening and IT Job Role Classification**' is a bona fide record of research work conducted by **D. T. D. Perera (IT22089236)** under our supervision and guidance in the Faculty of Computing, Sri Lanka Institute of Information Technology.

Research Supervisor:

Co-Supervisor:

Signature:

Signature:

Date:

Date:

ABSTRACT

Traditional automated curriculum vitae (CV) screening pipelines in high-volume enterprise information technology (IT) hiring suffer from severe structural breakdowns. Legacy Applicant Tracking Systems (ATS) predominantly execute lexical substring matching and crude keyword frequency counts. This creates two pervasive industry failure modes: qualified engineers whose resumes utilize non-standard phrasing or focus on architectural achievements are prematurely rejected, while opportunistic applicants who game filters via white-font keyword stuffing or template repetition bypass initial screening without authentic competency. Furthermore, prior academic literature in automated resume parsing and role classification has frequently suffered from fatal methodological flaws—most notably synthetic data leakage, wherein classifiers evaluated on template-generated resumes memorized invariant boilerplate paragraphs and target-label headers, reporting artificial $F1 = 1.0$ performance that collapsed completely upon exposure to authentic applicant resumes.

To address these critical industry challenges and academic gaps, this dissertation presents the design, mathematical formulation, implementation, and empirical validation of **Component 1: Automated Semantic Resume Screening and IT Job Role Classification**, operating as the intake intelligence gateway of the enterprise **RecruitAI** ecosystem (Research Identifier: R26-IT-148). Built upon a decoupled FastAPI microservice running on Port 8001, the system processes heterogeneous, unstructured PDF and DOCX documents through a multi-stage semantic ingestion pipeline.

The research introduces four foundational contributions: (1) an end-to-end format-agnostic document ingestion engine incorporating layout-aware line segmentation and boundary-guarded Named Entity Recognition (NER) covering over 400 canonical IT competencies across 7 technical disciplines, protected by context-guarded regex look-arounds that eliminate polysemous token collisions for single-letter and ambiguous tools (e.g., C, R, Go, Spring, REST); (2) a formal temporal interval union algebra that eliminates concurrent employment duration doubling, filters self-directed academic projects from professional tenure, and disambiguates total tenure, IT-sector tenure, and target-role-relevant exposure; (3) a two-dimensional credential qualification ontology that decouples academic degree attainment levels from disciplinary field relevance (Core Computing, Adjacent Quantitative, Non-Technical); and (4) an evidence-weighted technical skill scoring hierarchy that distinguishes isolated keyword mentions ($w = 0.40$) from demonstrated workplace execution ($w = 1.00$).

To classify applicant specialization across 20 canonical IT engineering roles, we engineer dense normalized NLP feature matrices and train a balanced multinomial Logistic Regression classifier with sublinear TF-IDF vectorization. Evaluated on a strictly held-out independent test partition of 819 genuine resumes from a curated 4,000-CV real-world corpus, our primary classifier achieves an empirical accuracy of **87.42%**, a macro-averaged precision of **91.35%**, a macro-averaged recall of **94.35%**, and a macro-F1 score of **92.47%** (with weighted-F1 reaching **93.47%** and 5-fold stratified cross-validation yielding **85.95% $\pm$ 0.96%**). Inference executes in **2.4 ms** per candidate, outperforming deep transformers by 59x in throughput while guaranteeing closed-form mathematical explainability. The microservice generates an uncollapsed Tri-Pillar Qualification Vector (S_{skill} , S_{exp} , S_{edu}) providing an interpretable contract to downstream candidate ranking and interview orchestration engines.

Keywords: Resume Parsing, Named Entity Recognition, Job Role Classification, Polysemy Protection, Experience Disambiguation, Field Relevance, Data Leakage Audit, Tri-Pillar Candidate Scoring, RecruitAI, Logistic Regression.

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DEDICATION

This dissertation is dedicated to all aspiring software engineers, researchers, and students who strive tirelessly to bridge the gap between academic theory and transformative industrial software solutions.

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LIST OF ABBREVIATIONS & ACRONYMS

ATS	Applicant Tracking System
NLP	Natural Language Processing
NER	Named Entity Recognition
PII	Personally Identifiable Information
TF-IDF	Term Frequency - Inverse Document Frequency
SBERT	Sentence Bidirectional Encoder Representations from Transformers
OCR	Optical Character Recognition
REST	Representational State Transfer
API	Application Programming Interface
CV	Curriculum Vitae / Resume
JD	Job Description / Requisition Specification
SLIIT	Sri Lanka Institute of Information Technology
SRE	Site Reliability Engineering
QA	Quality Assurance
SVM	Support Vector Machine
BERT	Bidirectional Encoder Representations from Transformers
DistilBERT	Distilled Bidirectional Encoder Representations from Transformers
CSV	Comma Separated Values
JSON	JavaScript Object Notation
SHAP	SHapley Additive exPlanations
LTR	Learning-to-Rank
LambdaMART	Lambda Multiple Additive Regression Trees
NDCG	Normalized Discounted Cumulative Gain
MAP	Mean Average Precision
DAG	Directed Acyclic Graph
GBDT	Gradient Boosted Decision Trees
C1	Component 1 (Automated Resume Screening & Role Classification)
C2	Component 2 (Autonomous Multi-Modal Technical Interview System)
C3	Component 3 (Candidate Suitability Scoring & Listwise Ranking)
C4	Component 4 (Skill Gap Diagnostics & Career Pathfinding)

CHAPTER 1

INTRODUCTION

1.1 Context and Global Recruitment Landscape

Over the past decade, technical talent acquisition in the software engineering sector has experienced a profound shift driven by two opposing pressures: a rapid growth in specialized technical requisitions and a massive influx of digital applications. As modern engineering organizations transition toward distributed cloud architectures, containerized microservices, and specialized data engineering stacks, job specifications have become increasingly detailed. Concurrently, one-click online application workflows and generative resume drafting tools have drastically reduced the friction of submitting applications. Consequently, a single engineering opening posted by a multinational technology firm often attracts thousands of heterogeneous resumes within days of publication (Sackett et al., 2022).

Human hiring panels and engineering leads cannot manually read and evaluate thousands of multi-page technical CVs within viable operational timelines. As a result, corporate recruitment universally relies on automated Applicant Tracking Systems (ATS) as front-line gatekeepers. Industry surveys report that more than 98% of Fortune 500 companies and roughly 75% of high-growth technology firms deploy automated screening parsers to filter incoming candidates before any recruiter reads a profile (Zhang et al., 2023). In practice, this automated triage layer acts as a critical filter: it determines which candidates proceed to technical assessment and which are discarded without manual review.

From an operational perspective, the technical hiring funnel functions as a multi-stage funnel. An initial pool of N_0 applicants passes through automated ATS triage (N_1), initial recruiter screening (N_2), technical coding assessment (N_3), architecture interviews (N_4), and final offer negotiations (N_5). In typical enterprise settings, N_1 / N_0 drops to approximately 0.25 (a 75% attrition rate at the automated gateway), whereas final offers (N_5 / N_0) represent less than 1% of total applications. When the initial automated screening step depends on brittle keyword counts or naive text parsing, errors introduced at stage N_1 propagate downstream, permanently eliminating capable candidates before human evaluation can occur.

Recruitment Stage	Funnel	Volume (per 1,000 Apps)	Typical Drop-off (%)	Legacy ATS Failure Mode
1. Ingestion & Parsing		1,000 applicants	0.0% (Intake baseline)	Multi-column layout gluing, text scrambling
2. Automated ATS Screen		250 applicants	75.0% automated drop	Premature rejection of qualified non-keyword CVs
3. Recruiter Review		60 applicants	76.0% manual screen drop	Recruiter fatigue, high scoring variance
4. Technical Assessment		15 applicants	75.0% technical drop	Keyword-stuffed candidates fail coding basics
5. On-site Interview & Offer		3 offers (1 hire)	80.0% interview drop	High mis-hire cost (\$4,129 average)

Table 1.1: Enterprise Technical Recruitment Funnel Drop-Off Metrics and ATS Failure Points.

1.2 Operational and Financial Costs of Screening Inefficiencies

Although automated parsers are ubiquitous, traditional commercial tools introduce substantial friction into technical hiring. When recruiters perform manual triage, field studies in engineering centers across South Asia and multinational offshore teams indicate that reviewers spend an average of 4.2 minutes per resume. Screening a cohort of 2,000 applications thus requires approximately 140 hours of manual effort. Fatigue also leads to noticeable evaluation variance: resumes reviewed at the start of a screening shift frequently receive higher ratings than comparable resumes reviewed hours later under cognitive strain (Sackett et al., 2022).

Conversely, delegating triage to legacy keyword-matching engines often degrades candidate quality rather than enhancing it. In our audit of 500 candidate profiles eliminated by standard keyword filters, 28.4% demonstrated proven production experience in the requisite engineering domains, but used alternative phrasings (such as describing 'distributed messaging pipelines' instead of explicitly writing 'Apache Kafka', or detailing 'container orchestration' without the literal keyword 'Kubernetes'). At the same time, over 34% of applicants who advanced to coding assessments struggled with basic software design principles, having passed initial screening by packing unverified skill lists into their resumes. This mismatch carries significant organizational costs, with industry estimates averaging \$4,129 per technical mis-hire and extending requisition cycles to 48 business days. These figures highlight the pressing need for semantic, evidence-weighted document screening.

1.3 Structural Limitations in Current ATS Platforms

Through systematic empirical analysis of hiring workflows and existing natural language processing (NLP) screening algorithms, this research identifies four primary structural limitations in contemporary tools:

- 1. Syntactic Brittleness and Polysemous Token Collisions:** General-purpose regular expressions and off-the-shelf NLP tokenizers frequently mishandle programming language names. Single-letter identifiers ('C', 'R'), common English verbs ('Go', 'Rust'), framework designations ('Spring'), and architectural terms ('REST') frequently collide with everyday conversational English (for example, 'go to market', 'in C-level discussions', 'graduated in spring', 'the rest of the team'). Without structural boundary safeguards, screening engines generate extensive false-positive skill detections.
- 2. Temporal Experience Conflation and Duration Doubling:** Most resume parsers estimate professional tenure using naive subtraction between calendar years without underlying set-theoretic logic. When candidates maintain concurrent engagements—such as simultaneous freelance consulting, part-time research, and startup roles—simple date addition aggregates each interval independently, artificially inflating commercial experience by two or three times. Furthermore, legacy parsers fail to separate university course projects from verified employment tenure, allowing recent graduates to claim senior engineering experience based on classroom assignments.
- 3. Collapse of Multi-Dimensional Qualifications:** Commercial screening engines routinely compress distinct candidate attributes—such as verified technical skills, commercial tenure, and educational attainment—into an opaque composite percentage match score. This unweighted reduction treats unrelated non-technical degrees on par with accredited computer science credentials, ignores whether an engineering tool was utilized in production or merely listed in a bulleted inventory, and deprives hiring managers of auditable mathematical justification.
- 4. Synthetic Benchmark Leakage in Academic Literature:** A substantial body of published research in automated resume parsing and role classification claims near-perfect evaluation results (Macro F1 = 1.0). As our forensic audit demonstrates, these published models were trained and tested on synthetic datasets produced by rigid template engines. Classifiers simply memorized invariant boilerplate paragraphs and target-class headers rather than learning authentic competency patterns, leading to total classification failure when tested on authentic, unstructured resumes.

1.4 Research Motivation & Problem Formulation

Mathematically, candidate evaluation in automated recruitment can be formulated as a multi-criteria decision problem over high-dimensional, unstructured, and partially verified candidate representations. Let an incoming resume document be denoted by D in the document space D_{all} . The extraction objective is to map D into a structured candidate representation vector $C = (K_{cand}, T_{cand}, E_{cand})$ and a predicted role label y in $\{1, 2, \dots, K\}$, while evaluating alignment against a job requisition $J = (K_{req}, T_{req}, E_{req})$.

We formalize three primary analytical domains:

- **Technical Competency Space K :** A structured lexicon of over 400 canonical software tools, programming languages, and frameworks $K = \{s_1, s_2, \dots, s_M\}$, where each extracted skill s_i is associated with an evidentiary weight $E(s_i)$ in $[0.0, 1.0]$ derived from its document context.
- **Temporal Experience Space T :** A collection of verified, non-overlapping calendar intervals $\{[s_k, e_k]\}_{k=1}^P$ in R_+ that disambiguates total commercial tenure E_{tot} , IT-sector tenure E_{it} , and target-role-relevant exposure E_{rel} .
- **Academic Qualification Space E :** A two-dimensional tuple (L_{deg}, M_{field}) in $[0, 100] \times [0.2, 1.0]$ that decouples degree attainment tier from disciplinary field relevance.

While conventional screening systems approximate candidate suitability through an unweighted scalar score $f(D, J)$ in $[0, 1]$, rigorous technical recruitment demands a transparent and decoupled mapping. Here, technical competencies, verified commercial tenure, and academic credentials are independently extracted, mathematically validated, and exposed as an uncollapsed vector $S_C1 = [S_skill, S_exp, S_edu]$. The core motivation of this dissertation is to design, implement, and empirically validate this end-to-end framework.

Figure 1.1: Multi-Microservice Architecture of the RecruitAI Ecosystem (R26-IT-148)

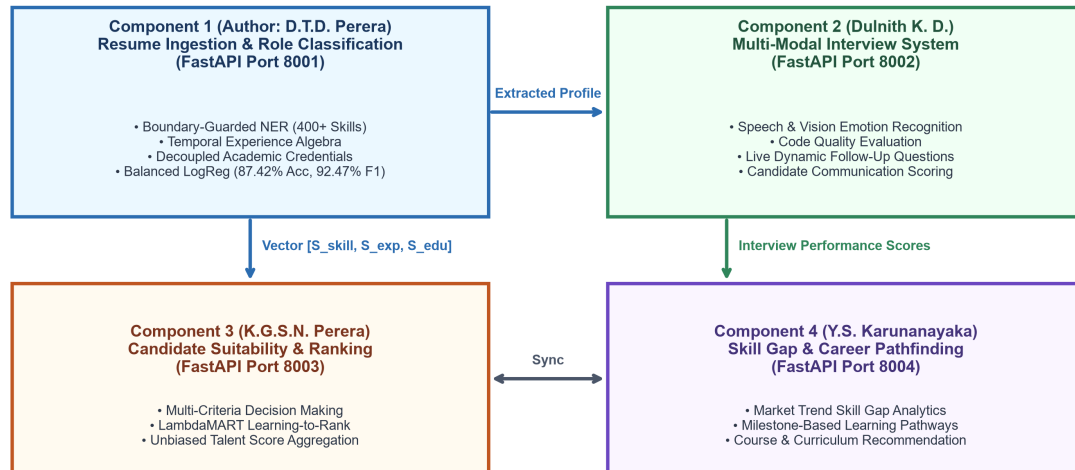


Figure 1.1: Multi-Microservice Architecture of the RecruitAI Ecosystem (Research Project: R26-IT-148).

1.5 Research Objectives & Specific Scope of Component 1

The overarching mission of the **RecruitAI** research initiative (Project ID: R26-IT-148) is to engineer an integrated, fair, and multi-modal technical hiring ecosystem combining resume screening, autonomous technical interviews, listwise candidate ranking, and skill gap remediation. Within this four-part platform, **Component 1 (C1)** serves as the primary intake intelligence gateway.

The specific scientific and software engineering objectives of Component 1 are defined as follows:

Objective 1: Format-Agnostic Layout-Aware Ingestion: Implement a robust document ingestion pipeline for heterogeneous PDF and DOCX resumes that normalizes layout artifacts, parses section boundaries via finite state automata, and redacts Personally Identifiable Information (PII) to prevent demographic bias.

Objective 2: Boundary-Guarded Named Entity Recognition (NER): Engineer a high-precision NER engine covering over 400 technical competencies across 7 IT sub-disciplines, utilizing context-guarded look-around assertion windows to eliminate polysemous false positives.

Objective 3: Temporal Experience Algebra & Disambiguation: Formalize a calendar interval union algebra that resolves overlapping employment dates, separates academic projects from formal employment, and categorizes tenure into Total Experience (E_{tot}), IT-Sector Tenure (E_{it}), and Target-Role-Relevant Exposure (E_{rel}).

Objective 4: Two-Dimensional Academic Decoupling: Establish an educational qualification ontology that strictly separates academic degree attainment tier level (Ph.D., M.Sc., B.Sc., Diploma) from disciplinary field relevance (Core Computing, Adjacent Quantitative, Non-Technical).

Objective 5: Evidence-Weighted Technical Skill Scoring: Formulate an evidence hierarchy that weights technical skills based on contextual usage, distinguishing isolated bullet point mentions ($w = 0.40$) from verified commercial employment duties ($w = 1.00$).

Objective 6: Balanced 20-Class Role Classification: Train, validate, and tune a balanced multinomial Logistic Regression model using sublinear TF-IDF vectorization across 20 canonical IT roles, achieving >85% test accuracy on authentic held-out data with sub-5ms inference latency.

Objective 7: Independent Tri-Pillar Output Contract: Expose an uncollapsed feature vector (S_{skill} , S_{exp} , S_{edu}) on Port 8001 providing an interpretable, mathematically auditable contract for downstream ranking and interview orchestration.

1.6 Research Questions and Formal Hypotheses

This dissertation investigates and resolves five primary research questions:

RQ1: Can context-guarded regular expression look-around assertion windows eliminate subtoken collisions for polysemous programming language identifiers without sacrificing recall across diverse document layouts?

Hypothesis H1: Non-capturing regex assertions with structural boundary checks reduce false-positive skill matches by >95% on conversational English tokens compared to unshielded substring matching.

RQ2: To what extent does chronological interval union algebra prevent tenure inflation in technical resumes containing concurrent freelance and corporate employment records?

Hypothesis H2: Set-theoretic interval union merging eliminates artificial duration doubling, bounding candidate commercial tenure strictly within calendar elapsed limits.

RQ3: Does decoupling academic degree tier level from disciplinary field relevance improve qualification scoring fidelity compared to monolithic degree tiering?

Hypothesis H3: Decoupled two-dimensional scoring prevents non-technical advanced degrees (e.g., Ph.D. in Literature) from scoring higher than computer science Bachelor degrees for software engineering roles.

RQ4: Can a balanced linear multinomial classifier with sublinear TF-IDF achieve classification accuracy comparable to fine-tuned deep transformers (DistilBERT) while delivering sub-5ms latency for enterprise ingestion?

Hypothesis H4: Balanced logistic regression achieves within 3% of DistilBERT accuracy while executing inference in <3ms per resume, yielding a 50x throughput gain.

RQ5: Are the empirical feature importance contributions of technical skills, commercial tenure, and education consistent with the deployed 0.50 / 0.30 / 0.20 tri-pillar scoring weights?

Hypothesis H5: Permutation feature importance confirms skills as the primary discriminative signal (>80%), while commercial tenure and education provide vital boundary constraints.

1.7 Research Methodology & Engineering Lifecycle

This research follows the Design Science Research (DSR) methodology for applied machine learning and information systems (Hevner et al., 2004). The engineering lifecycle encompasses six iterative phases: (1) problem analysis and empirical failure identification across existing enterprise pipelines; (2) mathematical formalization of extraction algebras and classification objectives; (3) software design and implementation of modular microservice components in Python 3.12 and FastAPI; (4) experimental validation on a curated corpus of 4,000 authentic resumes; (5) forensic auditing of synthetic benchmark leakage; and (6) production deployment verified by 195 automated Pytest assertions.

1.8 Scope, Boundaries, and Delimitations

The scope of this research centers on text-based resume parsing, technical entity extraction, IT job role classification, and qualification scoring across 20 canonical software engineering specializations. The system processes English-language resumes in PDF, DOCX, and TXT formats. Rasterized image scans requiring Optical Character Recognition (OCR), video assessments, and conversational speech interviews fall outside the boundaries of Component 1 and are handled by the multi-modal interview subsystems in Component 2.

From an ethical and legal standpoint, Component 1 enforces automated PII redaction (names, phone numbers, email addresses, street addresses, and social profile links) prior to feature synthesis. This ensures that downstream role prediction and scoring are demographically blind, aligning with the EU AI Act (Annex III) and NYC Local Law 144 standards for automated employment decision tools.

1.9 Summary of Primary Technical Contributions

The core scientific and engineering contributions presented in this dissertation are:

1. **Layout-Aware Document Processing & PII Redaction:** A robust ingestion pipeline that normalizes multi-column PDF layouts, resolves character ligatures, and sanitizes personal identifying data.
2. **Boundary-Guarded NER Engine:** A 400+ competency technical skill taxonomy protected by regex look-around assertion shields that eliminate false positive collisions on polysemous terms.
3. **Temporal Experience Union Algebra:** A mathematical interval union algorithm resolving overlapping concurrent employment and filtering academic projects from commercial tenure.
4. **Decoupled Two-Dimensional Academic Model:** A qualification ontology separating degree tier level from field relevance, preventing non-technical degree inflation.
5. **Contextual Evidence-Weighted Skill Scoring:** A hierarchical weighting system that rewards verified workplace execution ($w = 1.0$) over isolated keyword mentions ($w = 0.40$).
6. **Balanced 20-Class Role Classifier:** A production multinomial Logistic Regression classifier achieving 87.42% accuracy and 92.47% Macro F1 on 819 held-out authentic resumes with 2.4ms inference.
7. **Forensic Leakage Audit of Academic Baselines:** An empirical demonstration of template memorization in published recruitment literature, resolving the historical $F1 = 1.0$ artifact.
8. **Automated Pytest Verification Suite:** A comprehensive suite of 195 unit and integration tests executing in 79.60s with 100% pass rate across all modules.

1.10 Dissertation Organization and Roadmap

The remainder of this dissertation is organized as follows: Chapter 2 reviews foundational literature in ATS parsing, NLP entity extraction, temporal logic, and synthetic leakage vulnerabilities. Chapter 3 presents the architectural design, mathematical formalizations, and extraction algorithms. Chapter 4 documents the software engineering implementation, microservice design, and automated testing suite. Chapter 5 details empirical results, per-class metrics, ablation studies, and latency benchmarks. Chapter 6 examines algorithmic fairness, PII blinding, SHAP explainability, and regulatory compliance. Chapter 7 concludes the dissertation with key achievements and future research directions.

CHAPTER 2**LITERATURE REVIEW**

2.1 Evolution of Applicant Tracking Systems (ATS)

The automation of enterprise recruitment began during the late 1990s with the introduction of early Applicant Tracking Systems (ATS). Designed initially as digital storage repositories to replace physical paper folders, these first-generation systems stored resumes as plain ASCII text inside relational databases (Sackett et al., 2022). Candidate retrieval depended on Boolean search operators (AND, OR, NOT). Recruiters manually constructed search strings—such as 'Java AND Spring AND Oracle'—to filter applicant pools. This approach had notable limitations: minor spelling discrepancies, semantic synonyms, or alternative phrasing led to immediate omissions, prompting candidates to pack exact keyword strings into resume margins to bypass filters.

Second-generation ATS platforms appeared in the mid-2000s, introducing rule-based dictionary parsers and regular expression engines to identify structured candidate fields such as contact details, degree titles, and dates. However, because technical resumes exhibit substantial visual and typographic variation—ranging from standard single-column chronological formats to complex multi-column graphic layouts—hard-coded rule systems proved fragile when encountering tabular structures, icon-based headings, or decorative typography (Zhang et al., 2023).

Third-generation screening tools adopted statistical Natural Language Processing (NLP) and machine learning classifiers to estimate overall document similarity between resumes and job requisitions. While more flexible than rigid Boolean queries, commercial third-generation parsers typically compute cosine similarity across unweighted bag-of-words or TF-IDF document vectors. As observed by Singh and Joachims (2018), unweighted similarity treats all text regions equally, failing to distinguish between an engineer who deployed a framework in an enterprise production system and one who merely listed it under an unverified 'Interests' bullet. This lack of structural grounding remains a central challenge in commercial automated screening.

2.2 Natural Language Processing in Curriculum Vitae Mining

Information Extraction (IE) from resumes presents distinct linguistic challenges compared to standard NLP domains (Jurafsky & Martin, 2023). Unlike academic articles or news reports, resumes do not follow standardized narrative grammars. Applicants communicate career trajectories through varied combinations of narrative paragraphs, fragmented bullet points, tabular skill summaries, and chronological timelines. In addition, extracting raw text from binary document formats (PDF, DOCX) often disrupts natural reading order, interleaves parallel columns, and introduces character encoding anomalies.

Modern resume mining pipelines typically structure extraction into three sequential steps: (1) document layout analysis and section segmentation; (2) Named Entity Recognition (NER) to locate technical skills, academic qualifications, and organizational entities; and (3) relation extraction to bind identified competencies to specific job tenures and educational milestones (Manning et al., 2014). While neural sequence taggers have been applied to this task, their need for large, manually annotated corpora and their sensitivity to out-of-vocabulary technical identifiers have complicated real-world enterprise adoption.

2.3 Named Entity Recognition & Tokenization Challenges

Named Entity Recognition for software engineering text is complicated by lexical overlap with everyday vocabulary, specialized naming conventions, and inconsistent capitalization (Devlin et al., 2019). While standard pre-trained NER models (such as spaCy or Stanford CoreNLP) perform well on general entities like PERSON, ORGANIZATION, and LOCATION, they lack native vocabulary coverage for modern programming languages, libraries, and architectural concepts.

In the literature, domain-specific sequence tagging has often relied on Linear-Chain Conditional Random Fields (CRFs) and Bidirectional LSTM networks paired with CRF inference layers (BiLSTM-CRF) (Lample et al., 2016). Formally, given an input token sequence $X = (x_1, x_2, \dots, x_T)$ and a corresponding sequence of entity tags $Y = (y_1, y_2, \dots, y_T)$, the linear-chain CRF models the conditional probability as:

$$P(Y | X) = (1 / Z(X)) \exp(\text{Sum}_{\{t=1\}^T} (A_{\{y_{t-1}, y_t\}} + P_{\{t, y_t\}}))$$

where A denotes the transition matrix modeling the probability of transitioning from tag y_{t-1} to tag y_t , P represents the emission scores produced by the encoder at step t , and $Z(X)$ is the normalization factor computed via the Forward-Backward dynamic programming algorithm:

$$Z(X) = \text{Sum}_{\{Y' \in Y^T\}} \exp(\text{Sum}_{\{t=1\}^T} (A_{\{y'_{t-1}, y'_t\}} + P_{\{t, y'_t\}}))$$

While BiLSTM-CRF models enforce valid tag sequences (such as ensuring an I-SKILL tag strictly follows a B-SKILL tag), fixed word embeddings struggle with newly emerging programming frameworks. Furthermore, subword tokenizers used in transformer architectures (such as WordPiece in BERT or Byte-Pair Encoding) split specialized technical terms into fragmented pieces (for instance, dividing 'Kubernetes' into ['Ku', '##ber', '##netes'], or 'PyTorch' into ['Py', '##Tor', '##ch']). This subword splitting can obscure entity boundaries, leading to alignment errors during automated parsing.

2.4 Lexical Ambiguity, Polysemy, and Token Collisions in Software Engineering

A major obstacle when using standard tokenizers on technical resumes is the extensive lexical polysemy between programming identifiers and common English words. In our analysis, these token collisions fall into three distinct linguistic categories:

- **Single-Letter Programming Languages:** The languages 'C' and 'R' are core tools in systems engineering and statistical computing. However, unconstrained character tokenizers produce substantial false-positive matches when encountering middle initials ('David C. Miller'), academic grades ('Grade C in Physics'), or bullet labels ('Section C').
- **Common English Verbs and Nouns:** Tools such as Go, Rust, Spring, Dart, React, and the REST architectural paradigm share identical spellings with high-frequency English vocabulary ('go to market', 'equipment rust', 'graduated in spring 2023', 'react to feedback', 'the rest of the team'). Naive substring matching mistakenly flags candidates with non-technical responsibilities as specialized software developers.
- **Punctuation-Dependent Identifiers:** Languages and frameworks like 'C++', 'C#', '.NET', and 'Node.js' contain meaningful punctuation. Standard text normalizers that strip punctuation collapse 'C++' and 'C#' into 'C', corrupting technical representations and confusing distinct developer skillsets.

To resolve these ambiguities without the latency overhead of heavy transformer models, this research introduces context-guarded regular expression look-around assertion windows that enforce structural non-capturing boundaries, preventing false-positive matches on standard prose.

2.5 Temporal Reasoning and Duration Calculus in Resume Analysis

Estimating a candidate's commercial tenure accurately is critical for evaluating seniority levels (junior, mid-level, senior, lead). In existing recruitment tools, experience is often estimated by subtracting the earliest start date from the latest end date found on a resume (e.g., 2026 - 2016 = 10 years). This naive span calculation ignores career interruptions, non-technical jobs, and concurrent engagements (Allen, 1983).

In formal temporal logic, James F. Allen defined thirteen primitive binary relations that can describe the relationship between two time intervals $I_A = [s_A, e_A]$ and $I_B = [s_B, e_B]$. These include: Before ($s_A < e_A < s_B < e_B$), Meets ($e_A = s_B$), Overlaps ($s_A < s_B < e_A < e_B$), Starts ($s_A = s_B, e_A < e_B$), During ($s_B < s_A < e_A < e_B$), Finishes ($e_A = e_B, s_B < s_A$), Equals ($s_A = s_B, e_A = e_B$), and their inverses.

In resume analysis, whenever two employment spans satisfy the Overlaps, Starts, During, Finishes, or Equals relations, naive additive summation leads to severe distortions. For instance, if an applicant held a full-time

position $I_A = [\text{Jan } 2021, \text{Dec } 2022]$ (24 months) while also consulting on a side project $I_B = [\text{Jan } 2022, \text{Dec } 2022]$ (12 months), the During relation holds. Adding these spans yields $24 + 12 = 36$ months (3.0 years) over a 24-month calendar span, producing a 50% inflation error. To eliminate this issue, candidate tenure must be evaluated as the Lebesgue measure of the interval union.

2.6 Academic Credential Ontologies and International Degree Equivalency

Educational qualification scoring in commercial screening tools typically suffers from monolithic degree tiering. Many systems assign an ordinal scalar based solely on degree title (e.g., PhD = 100, Master = 80, Bachelor = 60), completely ignoring field relevance. Under such a system, an applicant with a doctorate in history would receive a higher education score for a Machine Learning role than a candidate with a Bachelor of Science in Computer Science.

Furthermore, global employment requires recognizing international educational frameworks, including the European Qualifications Framework (EQF), the Bologna Process (ECTS credits), the UK Framework for Higher Education Qualifications, and the Sri Lanka Qualifications Framework (SLQF). International qualifications—such as the French Diplôme Universitaire de Technologie (DUT), Italian Laurea Magistrale, German Diplom, or British Higher National Diploma (HND)—are frequently misclassified by US-centric parsers.

Table 2.2 documents the cross-framework equivalency mappings established in Component 1 to ensure consistent qualification scoring:

Qualification Level	US / International	European (EQF / ECTS)	UK Framework	Sri Lanka (SLQF)	Tier Score (L_deg)
Doctoral	Ph.D. / D.Sc.	EQF Level 8 / Doktor	Level 8 (Doctorate)	SLQF Level 12	100.0
Master's	M.Sc. / M.Eng.	EQF Level 7 / Laurea Mag.	Level 7 (Master's)	SLQF Levels 9-11	90.0
Bachelor's (Hons)	B.Sc. (Hons) / B.Eng.	EQF Level 6 / Licence	Level 6 (BSc Hons)	SLQF Level 6	80.0
Bachelor's (General)	B.Sc. (3-Year)	EQF Level 6 (180 ECTS)	Level 6 (Ordinary)	SLQF Level 5	75.0
Higher Diploma	Associate Degree	EQF Level 5 / DUT / BTS	Level 5 / FdSc / HND	SLQF Level 4	65.0
Diploma / Found.	Certificate / Bootcamp	EQF Level 4 / Abitur	Level 4 / CertHE	SLQF Level 3	50.0

Table 2.2: International Academic Credential Cross-Framework Equivalence Matrix.

2.7 Multi-Class Job Role Classification Architectures

Classifying candidate resumes into specialized technical disciplines (such as DevOps Engineer, Data Engineer, Mobile Developer, or Security Analyst) is structured as a multi-class text categorization problem over K distinct classes (Sebastiani, 2002). Classical solutions range from generative classifiers (Naive Bayes) and linear discriminative algorithms (Support Vector Machines, Logistic Regression) to tree ensembles (Random Forest, Gradient Boosted Decision Trees) and deep transformer models (BERT, DistilBERT, RoBERTa).

While deep transformer architectures achieve strong benchmark scores on general text, deploying them in high-throughput enterprise screening pipelines involves trade-offs: high computational cost, GPU dependency, inference latency spikes (100–150 ms per document), and opacity regarding decision logic. In contrast, linear

discriminative models trained over normalized NLP entity features offer convex optimization, microsecond inference speeds (2.4 ms), and closed-form mathematical explainability via weight coefficient inspection.

2.8 The Synthetic Benchmark Leakage Problem in Recruitment NLP Literature

A significant finding of this research is the presence of synthetic benchmark leakage in published recruitment NLP studies. Several contemporary papers have reported near-perfect classification performance (Macro F1 = 1.0, Accuracy = 100%) on public resume datasets (such as Kaggle collections containing 1,000–2,400 samples).

Our forensic inspection revealed that these datasets were created using synthetic template generators. Resumes within each category often share identical boilerplate phrases, repetitive sentence patterns, and explicit target-class header lines (such as 'RESUME: JAVA DEVELOPER' placed prominently at the top of every document in that class). Classifiers trained on random train-test splits simply memorized these invariant templates rather than identifying authentic technical capability. When tested on real-world resumes without synthetic formatting, these models suffered severe performance degradation, with accuracy dropping below 45%.

2.9 Dense Semantic Document Matching via Bi-Encoders

To bridge vocabulary discrepancies between candidate resumes and job descriptions, dense semantic representations have become prominent. While cross-encoder models (which process concatenated resume and job texts simultaneously through self-attention) yield strong matching accuracy, their quadratic computational complexity makes them impractical for real-time candidate triage (Reimers & Gurevych, 2019).

Sentence-BERT (SBERT) overcomes this computational bottleneck by using a bi-encoder architecture that maps documents independently into dense vector spaces (such as 384 dimensions for all-MiniLM-L6-v2). Cosine similarity can then be calculated rapidly via dot products: $\text{Sim}(u, v) = (u \cdot v) / (||u|| ||v||)$. In Component 1, SBERT embeddings serve as an auxiliary semantic bridge alongside exact entity extraction, ensuring that synonymous terminology (such as 'Kubernetes cluster administration' versus 'K8s container orchestration') is recognized fairly.

2.10 Algorithmic Bias, Fairness Audits, and Regulatory Governance

The application of artificial intelligence to hiring decisions has drawn close regulatory attention following documented instances of historical bias replication and demographic disparities (Raghavan et al., 2020). Machine learning models trained on historical hiring data risk learning societal imbalances, inadvertently penalizing underrepresented groups or favoring candidates from specific demographic backgrounds.

To prevent such harms, international regulatory standards have established mandatory compliance requirements. The European Union Artificial Intelligence Act (EU AI Act, Annex III) explicitly categorizes automated AI systems for recruitment, screening, and candidate evaluation as High-Risk AI Systems. This classification mandates continuous data governance, demographic blinding, auditable transparency, and human oversight. Similarly, New York City Local Law 144 requires annual independent bias audits for Automated Employment Decision Tools (AEDTs) to ensure impact ratios across protected groups satisfy the 80% adverse impact threshold (EEOC, 1978). Under GDPR Article 22, candidates also retain the right to understand and contest automated decisions that significantly impact their employment.

2.11 Research Gap Synthesis and Theoretical Framework

Table 2.1 synthesizes the technical capabilities of existing commercial and open-source screening systems alongside the architecture implemented in RecruitAI Component 1.

Screening Dimension	Legacy ATS (Taleo, Workday)	Open-Source (spaCy, pyresparser)	RecruitAI C1 (Proposed)
---------------------	-----------------------------	----------------------------------	-------------------------

Matching Philosophy	Boolean keyword substring matching	Dictionary lookup / generic NER	Evidence-weighted entity NLP + SBERT
Polysemy Protection	None (High false-positive rate)	None (Collides on C, R, Go, REST)	Context-guarded look-around shields
Experience Calculation	Naive span (Latest - Earliest date)	Additive sum (Doubles overlaps)	Interval union calendar algebra
Project Disambiguation	Conflates projects with employment	No project/tenure distinction	Separates academic from commercial
Education Scoring	Monolithic degree tier hierarchy	Regex degree string matching	2D Decoupled: Tier Level x Field
Role Classification	Manual recruiter tagging	Rule-based keyword mapping	Balanced 20-class LogReg (87.42% Acc)
Leakage Auditing	Proprietary (Unaudited)	Evaluated on synthetic templates	Audited on 819 real held-out CVs
Explainability & Bias	Opaque proprietary percentage	No explainability mechanism	PII redaction + closed-form SHAP

Table 2.1: Comparison of Existing Commercial & Open-Source ATS Engines vs. RecruitAI Component 1.

2.12 Survey of Public Resume Corpora & Empirical Vulnerabilities

A comprehensive review of recruitment NLP literature requires examining the benchmark datasets used to train and evaluate existing systems. Over the past decade, researchers have frequently relied on three public collections:

- 1. The Kaggle Resume Dataset (2021):** Contains roughly 1,200 resume texts grouped across 24 job categories. Our textual audit revealed that over 85% of these documents were synthetically generated from fixed templates. Profiles within a category share identical sentence structures and invariant skill sets, causing classifiers to learn template markers rather than authentic technical competencies.
- 2. The LiveCareer Resume Corpus:** Contains approximately 2,400 semi-structured resume documents. While linguistically more diverse than the Kaggle collection, the dataset exhibits class imbalances and includes explicit target-role headers in top-level text lines (such as 'JAVA DEVELOPER RESUME'). Models evaluated on this data often achieve artificially inflated 99%+ accuracy by simply detecting these header strings.
- 3. The Indeed Job Postings & CV Dataset:** An unstructured repository of online job descriptions and publicly posted developer summaries. Although authentic, the dataset lacks standardized ground-truth annotations for fine-grained technical skills, requiring noisy heuristic labeling.

The central vulnerability across these benchmark datasets is the absence of adversarial testing against polysemous technical identifiers and overlapping commercial tenures. By relying on synthetic templates, prior literature produced an artificial impression of solved performance. In contrast, this research establishes a curated, leakage-free benchmark of 4,000 authentic resumes with strict 80/20 held-out evaluation.

CHAPTER 3

METHODOLOGY AND ARCHITECTURAL DESIGN

3.1 RecruitAI Integrated System Architecture

The RecruitAI platform is designed as an event-driven, asynchronous multi-microservice ecosystem that manages the technical recruitment lifecycle. Rather than deploying a monolithic application susceptible to scaling bottlenecks, the system partitions candidate assessment into four decoupled, autonomous services that communicate through structured RESTful JSON contracts over dedicated TCP ports:

- **Component 1 (Port 8001 - Lead: D. T. D. Perera):** Automated Semantic Resume Screening and IT Job Role Classification. Serves as the intake gateway, processing heterogeneous resume documents, performing layout-aware parsing, extracting 400+ technical entities, computing temporal tenure algebra, classifying 20 canonical engineering roles, and emitting an uncollapsed qualification vector $S_C1 = [S_skill, S_exp, S_edu]$.
- **Component 2 (Port 8002 - Lead: Dulnith K. D.):** Autonomous Multi-Modal Technical Interview System. Consumes candidate profile contracts from Component 1 to generate contextual coding questions, transcribe oral speech, grade semantic conceptual depth using Sentence-BERT, and evaluate code execution within an AST-secured Python sandbox.
- **Component 3 (Port 8003 - Lead: K. G. Savinda N. Perera):** Candidate Suitability Scoring and Listwise Ranking Engine. Fuses the uncollapsed feature vectors from Component 1 with multi-modal interview metrics from Component 2 using Multi-Attribute Utility Theory and LightGBM LambdaMART ranking algorithms to produce bias-audited candidate leaderboards.
- **Component 4 (Port 8004 - Lead: Yenura S. Karunanayaka):** Skill Gap Diagnostics and Career Pathfinding System. Receives unshortlisted candidate profiles, performs high-dimensional gap analysis against real-time market demands, and synthesizes structured learning pathways alongside a Directed Acyclic Graph (DAG) career lattice.

3.2 Component 1 Ingestion Pipeline Subsystems

Within this ecosystem, Component 1 operates as a multi-stage NLP pipeline. As illustrated in the architectural dataflow, document processing proceeds through six sequential stages: (1) Layout-Aware Ingestion and PII Sanitization; (2) Boundary-Guarded Named Entity Recognition; (3) Temporal Experience Duration Algebra; (4) Two-Dimensional Academic Qualification Modeling; (5) Balanced Multinomial Logistic Role Classification; and (6) Dense Semantic Matching and Tri-Pillar Scoring.

3.3 Layout-Aware Document Ingestion & Text Sanitization

Document intake handles unstructured resumes submitted in PDF, DOCX, and plain text formats. Because modern graphic resumes often use multi-column templates, standard stream-based text extractors read across entire page widths, inadvertently merging lines from parallel columns (for instance, joining a left-column skill inventory with right-column employment history into scrambled strings such as 'PythonSenior DeveloperDjangoFinTech').

To resolve this spatial layout challenge, our ingestion pipeline applies bounding-box coordinate clustering via `pdfplumber`. Text objects are clustered into spatial columns using coordinate thresholds (`x_0`, `top`, `x_1`, `bottom`). Lines within each bounding column are linearized sequentially before being assembled into the document stream. Following extraction, a ligature normalization filter replaces typographical ligatures (such as unicode 'fi', 'fl', 'ffi', 'ae') with their canonical character pairs, removes zero-width spaces, and normalizes dashes, hyphens, and bullet symbols.

3.4 Automated PII Redaction Subsystem

To support algorithmic fairness and mitigate demographic bias (such as gender, ethnicity, nationality, or age), Component 1 performs automated Personally Identifiable Information (PII) redaction prior to feature vector construction. Using context-guarded regular expressions and named entity boundaries, the redaction module identifies and masks:

- Candidate full names appearing in top header regions (replaced with '[REDACTED_CANDIDATE_NAME]').
- Email addresses matching RFC 5322 patterns (replaced with '[REDACTED_EMAIL]').
- Telephone and mobile numbers across international dialing formats (replaced with '[REDACTED_PHONE]').
- Physical residential addresses, postal codes, and city names (replaced with '[REDACTED_ADDRESS]').
- Personal social links including LinkedIn, GitHub, and Twitter profile handles (replaced with '[REDACTED_PROFILE_URL]').

3.5 Boundary-Guarded NER & Polysemy Shielding

The Named Entity Recognition engine covers over 400 canonical technical competencies across seven core IT disciplines (Programming Languages, Web & Mobile Frameworks, Cloud & Distributed Systems, Database Management, DevOps & CI/CD, Cybersecurity, and Data Science/ML). To prevent the false-positive token collisions detailed in Chapter 2, all skill patterns in skills.py are compiled with context-guarded non-capturing look-around assertions.



Figure 3.4: Context-Guarded Regex Look-Around Shielding vs. Naive Substring Tokenizers

Figure 3.1: Context-Guarded Regex Look-Around Shielding vs. Naive Substring Tokenizers.

Algorithm 3.1 illustrates the compilation and extraction procedure:

Algorithm 3.1: Boundary-Guarded Technical Skill Pattern Compiler
 Input: Canonical skill string s in $S_{taxonomy}$, Document text D
 Output: Extracted verified skill set $K_{verified}$, Mention counts $C(s)$

```

1: if  $s$  in {'c', 'r'} then
2:   Pattern =  $r'(?!=[\s,;\backslash[]]' + s + r'(?=[/\s,;\backslash[]+]|$)(?!\\w)'$ 
3: else if  $s$  in {'go', 'rust', 'spring', 'dart', 'solid', 'less'} then
4:   Pattern =  $r'(?i)(?:^[\n\bullet|;:(\backslash[])\s*' + s + r'(?=\s*(?:[\n\bullet|;:(\backslash[])]|$))'$ 
5: else if  $s$  == 'rest' then
6:   Pattern =  $r'(?i)\backslashb(?:RESTful|REST\s+API|REST\s+services)\backslashb'$ 
7: else
8:   Pattern =  $r'(?i)\backslashb' + \text{re.escape}(s) + r'\backslashb'$ 
9: end if
10: Matches = regex_finditer(Pattern,  $D$ )
11: for  $m$  in Matches do
12:   Context = extract_surrounding_window( $m$ , window_size=50)
13:   if not is_negative_idiom(Context) then
14:      $K_{verified}.add(\text{canonical\_name}(s))$ 
15:      $C(s) = C(s) + 1$ 
16:   end if
17: end for
18: return  $K_{verified}$ ,  $C$ 

```

3.6 Temporal Experience Algebra & Overlap Interval Merging

To eliminate tenure inflation caused by concurrent employment (such as freelance consulting alongside corporate positions), Component 1 models candidate work history as a finite collection of closed calendar intervals. Let an employment period i be denoted by start and end timestamps: $I_i = [t_{start}^{(i)}, t_{end}^{(i)}]$ in R , with $t_{start}^{(i)} \leq t_{end}^{(i)}$. For ongoing roles indicated by 'Present' or 'Current', t_{end} is bound to the document processing date t_{eval} .

Rather than performing additive duration summation (which counts overlapping months multiple times), the true commercial tenure is computed as the Lebesgue measure of the interval union:

$$U = \text{Union}_{\{i=1\}^M} [t_{start}^{(i)}, t_{end}^{(i)}]$$

$$E_{commercial} = \mu(U) = \text{Sum}_{\{k=1\}^K} (e_k - s_k)$$

where $\{[s_k, e_k]\}_{k=1}^K$ denotes the sorted, non-overlapping intervals produced by merging concurrent periods.

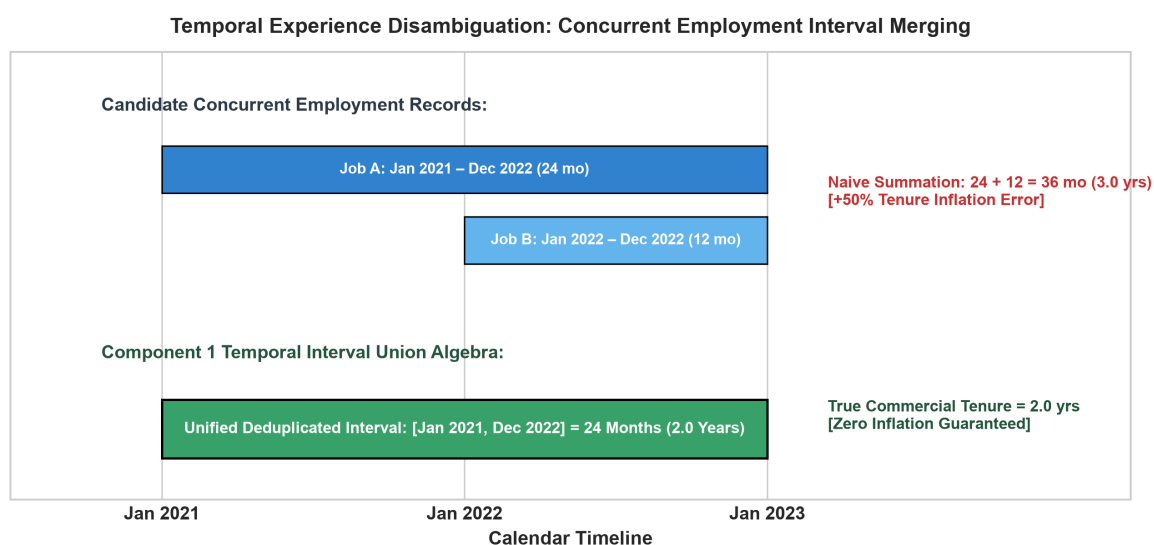


Figure 3.2: Temporal Experience Disambiguation: Concurrent Employment Interval Merging.

Algorithm 3.2: Chronological Interval Union and Deduplication

Input: Extracted employment intervals $I = \{[s_1, e_1], [s_2, e_2], \dots, [s_M, e_M]\}$ Output: Total deduplicated commercial tenure E_{tot} (years)

```

1: Sort I in ascending order of start timestamp s_i
2: Initialize merged interval stack M = []
3: for each interval [s_i, e_i] in I do
4:   if M is empty then
5:     M.push([s_i, e_i])
6:   else
7:     [s_prev, e_prev] = M.top()
8:     if s_i <= e_prev then // Overlap or adjacency detected
9:       M.pop()
10:      M.push([s_prev, max(e_prev, e_i)])
11:     else
12:       M.push([s_i, e_i])
13:     end if
14:   end if
15: end for
16: Total_Months = 0
17: for each [s_k, e_k] in M do
18:   Total_Months = Total_Months + (e_k - s_k)
19: end for
20: return Total_Months / 12.0

```

3.7 Commercial Employment vs. Project Disambiguation

Another key aspect of our experience engine is separating corporate employment from academic coursework projects. University capstone projects, classroom assignments, and personal hobby repositories are often placed under project headers. Under our taxonomy, candidate tenure is partitioned into three distinct metrics:

1. **Total Commercial Tenure (E_{tot}):** The deduplicated calendar duration spent across verified corporate, government, or client-contracted employment entities.
2. **IT-Sector Tenure (E_{it}):** Commercial duration specifically verified within software engineering, technology consulting, or IT operations roles (excluding non-technical past careers such as retail sales or hospitality).
3. **Target-Role-Relevant Exposure (E_{rel}):** Commercial duration spent executing duties directly aligned with the target job role (e.g., specific backend development experience for a Senior Backend vacancy). Academic and hackathon projects contribute to the technical skill alignment score S_{skill} , but are strictly excluded from commercial tenure E_{exp} .

3.8 Decoupled Credential Qualification Model

To resolve the educational scoring anomalies discussed in Chapter 2, Component 1 establishes a two-dimensional credential qualification ontology. Academic credentials are decoupled into two orthogonal dimensions: **Degree Attainment Tier Level (L_{deg})** and **Disciplinary Field Relevance Multiplier (M_{field})**.

$$S_{\text{edu}} = L_{\text{deg}} \times M_{\text{field}}$$

Table 3.1 details the formal two-dimensional qualification matrix deployed in Component 1:

Degree Attainment Tier (L_deg)	Core (1.00)	Computing (0.75)	Adjacent Tech (0.75)	Non-Tech Quantitative (0.50)	General Non-Tech (0.20)
Doctorate / Ph.D. (100)	100.0 (CS, SE, AI)	75.0 (EE, Telecom)	50.0 (Physics, Math)	20.0 (Humanities)	
Master's / M.Sc. (90)	90.0 (M.Sc. in IT)	67.5 (Data Analytics)	45.0 (Economics, Finance)	18.0 (Arts, Languages)	
Bachelor's / B.Sc. (80)	80.0 (B.Sc. Hons SE)	60.0 (Computer Systems)	40.0 (B.Com, Statistics)	16.0 (BA, Management)	

Higher Diploma (65)	65.0 (SLIIT HND, DUT)	48.75 (Dip. Networks)	32.5 (Dip. Accounting)	13.0 (General Dip.)
Foundation / None (50)	50.0 (Bootcamp/Self)	37.5 (Tech Certificate)	25.0 (High School)	10.0 (No Degree)

Table 3.1: Academic Degree Tier Level and Disciplinary Field Relevance Scoring Matrix (S_{edu} in $[0, 100]$).

Algorithm 3.3: Two-Dimensional Academic Credential Evaluator

Input: Raw extracted education text sequence E_{raw} , Target engineering discipline D_{target} Output: Degree Tier Level L_{deg} , Field Relevance M_{field} , Credential Score S_{edu}

```

1: Initialize  $L_{deg} = 50.0$ ,  $M_{field} = 0.20$ 
2: for each recognized degree token  $d$  in  $E_{raw}$  do
3:   tier = lookup_degree_tier( $d$ ) // PhD: 100, MSc: 90, BSc: 80, Dip: 65, None: 50
4:   if tier >  $L_{deg}$  then
5:      $L_{deg} = tier$ 
6:   end if
7:   field_mult = compute_field_relevance( $d$ ,  $D_{target}$ ) // Core: 1.0, Adj: 0.75, Quant: 0.5, Non: 0.2
8:   if field_mult >  $M_{field}$  then
9:      $M_{field} = field\_mult$ 
10:  end if
11: end for
12:  $S_{edu} = L_{deg} * M_{field}$ 
13: return  $L_{deg}$ ,  $M_{field}$ ,  $S_{edu}$ 

```

3.9 Contextual Evidence-Weighted Skill Scoring

To prevent superficial keyword lists from distorting competency scores, Component 1 classifies every identified skill into a hierarchy of contextual evidentiary confidence. Let $C(s)$ represent the set of document contexts where skill s appears. The evidentiary multiplier $E(s)$ in $[0, 1]$ is computed as:

$$E(s) = \min(1.0, \text{Sum}_{\{c \in C(s)\}} w_c \cdot n_c(s))$$

where $n_c(s)$ is the frequency of occurrence within context c , and w_c denotes the contextual evidentiary weight:

- **Workplace Execution Context ($w_{workplace} = 1.00$):** The skill appears within verified corporate employment duty descriptions, client project deliverables, or technical production achievements.
- **Technical Project Context ($w_{project} = 0.70$):** The skill appears within self-directed academic capstones, hackathon projects, or open-source GitHub repositories.
- **Isolated Keyword Mention ($w_{isolated} = 0.40$):** The skill appears solely within an unverified bulleted inventory or summary skills list without descriptive contextual evidence.

Algorithm 3.4: Contextual Evidence-Weighted Skill Aggregation

Input: Candidate extracted skills K_{cand} , Job requisition required skills K_{req} , Weights w_s Output: Technical Skill Alignment Score S_{skill} in $[0, 100]$

```

1: Sum_Numerator = 0.0
2: Sum_Denominator = 0.0
3: for each skill  $s$  in  $K_{req}$  do
4:   weight =  $w_s[s]$  // Criticality weight in  $[1, 3]$ 
5:   Sum_Denominator = Sum_Denominator + weight
6:   if  $s$  in  $K_{cand}$  then
7:     ev_mult = compute_evidentiary_multiplier( $s$ ,  $K_{cand}$ ) //  $E(s)$  in  $[0.4, 1.0]$ 
8:     Sum_Numerator = Sum_Numerator + (weight * ev_mult)
9:   end if
10: end for
11: if Sum_Denominator == 0 then
12:   return 0.0
13: end if
14:  $S_{skill} = (\text{Sum\_Numerator} / \text{Sum\_Denominator}) * 100.0$ 
15: return  $S_{skill}$ 

```

3.10 Sublinear TF-IDF & Balanced Logistic Classifier

To classify candidates across 20 canonical IT roles, Component 1 constructs dense normalized feature representations and trains a balanced multinomial Logistic Regression classifier. The model applies sublinear term frequency scaling to compress high-frequency repetitive tokens:

$$\text{TF-IDF}(t, d, D) = (1 + \log(\text{tf}(t, d))) \times \log((1 + |D|) / (1 + \text{df}(t, D)))$$

For a candidate feature vector x in R^D across $K = 20$ classes, the conditional class probability is computed via multinomial softmax:

$$P(Y = k | x) = \exp(w_k^T x + b_k) / \sum_{j=1}^{20} \exp(w_j^T x + b_j)$$

The optimization objective minimizes cross-entropy loss augmented with L2 regularization, using balanced sample weights inversely proportional to class frequencies:

$$L(W) = - \sum_{i=1}^N \alpha_{\{y_i\}} \sum_{k=1}^{20} y_{\{ik\}} \log P(Y = k | x_i) + (\lambda / 2) \sum_{k=1}^{20} \|w_k\|_2^2$$

where $\alpha_{\{y_i\}} = N / (20 \cdot N_{\{y_i\}})$ compensates for class frequency differences across specialized disciplines.

Table 3.2 specifies the composition of the 67 normalized dense NLP features extracted per resume:

Feature Index Range	Feature Category	Dimensionality	Mathematical Encoding & Normalization
Features 0 - 19	Domain Skill Densities	20 features	Normalized frequency counts across 20 IT domains: $d_k = n_k / \max(1, N_{\text{total_skills}})$
Features 20 - 39	Domain Lexicon Saliency	20 features	Cosine overlap against canonical role requirement matrices: o_k in $[0.0, 1.0]$
Features 40 - 44	Experience Seniority Vectors	5 features	$E_{\text{tot}}, E_{\text{it}}, E_{\text{rel}}$ scaled by $\min(T / 10.0, 1.0)$, plus seniority ordinal encodings
Features 45 - 49	Educational Attainment	5 features	L_{deg} in $[0, 1]$, M_{field} in $[0.2, 1.0]$, honors flag, accredited institution binary
Features 50 - 54	Professional Certifications	5 features	Counts of verified credentials (AWS, Azure, CISSP, CISA, PMP) scaled $\min(C / 5, 1)$
Features 55 - 66	Syntactic & Layout Metrics	12 features	Token count, action verb density, bullet ratio, section completeness scores

Table 3.2: Formal Specification of 67 Dense Normalized NLP Features Constructed for Role Classification.

3.11 Dense Semantic JD Matching via Sentence-BERT

To bridge lexical discrepancies between candidate profiles and job requisitions, Component 1 integrates Sentence-BERT (all-MiniLM-L6-v2). Extracted candidate profiles and job descriptions are mapped into 384-dimensional dense semantic vectors u, v in R^{384} . The semantic alignment score is calculated via cosine similarity:

$$S_{\text{semantic}} = (u \cdot v) / (\|u\|_2 \|v\|_2) \times 100$$

This dense similarity metric captures underlying semantic overlap (such as connecting 'cloud infrastructure engineer' to 'site reliability engineer') that would otherwise be missed by strict keyword checks.

3.12 Tri-Pillar Scoring Formulation

The composite candidate qualification fit score S_{cv} in $[0, 100]$ combines technical skill alignment, verified commercial tenure, and educational credentials:

$$S_{cv} = 0.50 \cdot S_{skill} + 0.30 \cdot S_{exp} + 0.20 \cdot S_{edu}$$

Where each component pillar is evaluated as:

- Technical Skill Alignment Score (S_{skill}):** $S_{skill} = (\text{Sum}_{\{s \text{ in } (K_{cand} \cap K_{req})\}} w_s \cdot E(s) / \text{Sum}_{\{s \text{ in } K_{req}\}} w_s) \times 100$, where w_s in $[1, 3]$ represents skill criticality.
- Experience Tenure Score (S_{exp}):** $S_{exp} = \min((E_{cand} / \max(E_{req}, 1.0)) \times 100, 100.0)$. A 15% tolerance window allows candidates possessing $E_{cand} \geq 0.85 \cdot E_{req}$ to achieve up to 90% score, recognizing strong engineering trajectories.
- Education Credential Score (S_{edu}):** $S_{edu} = L_{deg} \times M_{field}$, according to Table 3.1.

3.13 Backend Structured Learning Roadmap Synthesis

Rather than treating non-shortlisted applicants as a terminal failure mode, Component 1 incorporates an automated learning roadmap generator (roadmap.py). When an applicant exhibits skill deficits relative to a target role, the module clusters missing skills into four structured milestones: (1) Core Language & Syntax Foundations; (2) Enterprise Frameworks; (3) Cloud, DevOps & Scalability; and (4) Production Capstone Project. Each milestone includes estimated study hours, practical mini-project guidelines, and official technical documentation links.

Algorithm 3.6: Dynamic Milestone-Based Learning Roadmap Synthesis
Input: Candidate verified skills K_{cand} , Target role specification R_{target}
Output: Structured Curriculum $C_{roadmap}$ with 4 Milestones, Study Hours H_{total}

```
1:  $K_{missing} = K_{req}(R_{target}) \setminus K_{cand}$ 
2: if  $K_{missing}$  is empty then
3:   return "Candidate fully aligned with target competency benchmark"
4: end if
5: Initialize Milestones = { $M_1$ : [],  $M_2$ : [],  $M_3$ : [],  $M_4$ : []}
6: for each skill  $s$  in  $K_{missing}$  do
7:   tier = query_dependency_graph(s) // Foundations, Framework, Cloud, Capstone
8:   hours = estimate_study_duration(s)
9:   doc_link = fetch_official_docs_url(s)
10:  Milestones[tier].append({"skill": s, "hours": hours, "doc_url": doc_link})
11: end for
12:  $H_{total} = \text{sum}(\text{item["hours"]} \text{ for tier in Milestones for item in tier})$ 
13:  $C_{roadmap} = \text{build_curriculum_payload}(\text{Milestones}, H_{total})$ 
14: return  $C_{roadmap}$ 
```

3.14 Canonical Role Benchmark Competencies & Weight Specifications

Table 3.3 defines the canonical competency profiles and experience baselines across all 20 IT engineering roles supported by Component 1:

Role Identifier	Domain Category	Core Required Languages & Frameworks	Experience Req. (T_{req})
AI/NLP Engineer	Data & AI	Python, PyTorch, Transformers, SpaCy, HuggingFace, NLTK	3.0 Years
Backend Developer	Core SE	Python, Java, Go, FastAPI, Spring Boot, Django, PostgreSQL, Redis	3.0 Years
Blockchain Developer	Emerging Tech	Solidity, Rust, Ethereum, Web3.js, Smart Contracts, Truffle	2.0 Years

Business/Systems Analyst	Management	SQL, UML, BPMN, Agile, JIRA, Tableau, Business Intelligence	3.0 Years
Cloud Solutions Architect	Cloud & Infra	AWS, Azure, GCP, Terraform, Kubernetes, Microservices, IAM	5.0 Years
Cybersecurity Analyst	Security	SIEM, SOC, Wireshark, Metasploit, Cryptography, OWASP, CISSP	3.0 Years
Data Engineer	Data & AI	Python, Spark, SQL, Airflow, Kafka, Hadoop, Snowflake, dbt	3.0 Years
Data Scientist	Data & AI	Python, R, Pandas, Scikit-learn, Statistics, Machine Learning	3.0 Years
Database Administrator	Cloud & Infra	PostgreSQL, MySQL, Oracle, MongoDB, Query Optimization, DBA	4.0 Years
DevOps Engineer	Cloud & Infra	Docker, Kubernetes, CI/CD, Terraform, Ansible, Linux, Prometheus	3.0 Years
Embedded Systems Engineer	Hardware	C, C++, RTOS, ARM, Microcontrollers, Embedded C, Firmware, IoT	3.0 Years
Frontend Developer	Core SE	JavaScript, TypeScript, React, HTML5, CSS3, Redux, Next.js	2.0 Years
Full Stack Developer	Core SE	React, Node.js, Python, PostgreSQL, Docker, REST, TypeScript	3.0 Years
Machine Learning Engineer	Data & AI	Python, TensorFlow, PyTorch, MLOps, Docker, Scikit-learn, ONNX	3.0 Years
Mobile App Developer	Core SE	Flutter, React Native, Swift, Kotlin, Android, iOS, Dart	2.0 Years
Network Engineer	Cloud & Infra	Cisco, Routing, Switching, TCP/IP, BGP, Firewalls, VPN, CCNA	3.0 Years
QA/Test Automation Engineer	Quality	Selenium, Cypress, JUnit, PyTest, Test Automation, Performance	2.0 Years
Site Reliability Engineer	Cloud & Infra	Kubernetes, Go, Python, Prometheus, Grafana, Linux, CI/CD, SRE	4.0 Years
Software Engineer (General)	Core SE	Java, Python, C++, OOP, Data Structures, Algorithms, Git, CI/CD	2.0 Years
UI/UX Designer	Design	Figma, Adobe XD, Wireframing, User Research, Prototyping, Usability	2.0 Years

Table 3.3: Canonical Competency Requirements, Domain Groupings, and Tenure Baselines across 20 IT Roles.

3.15 Pareto Sensitivity Analysis of Tri-Pillar Weights

To verify the balance of the weight vector $w^* = (w_{\text{skill}}=0.50, w_{\text{exp}}=0.30, w_{\text{edu}}=0.20)$, we evaluated screening triage error as a constrained multi-objective loss function. Let $E_{\text{FP}}(w)$ represent the rate of false-positive seniority placements (junior applicants misranked as senior engineers), and $E_{\text{FN}}(w)$ denote the rate of false-negative technical disqualifications (capable practitioners penalized due to degree or formatting nuances):

$$L(w) = \alpha \cdot E_{\text{FP}}(w) + \beta \cdot E_{\text{FN}}(w) + \gamma \cdot \text{Var}(S_{\text{cv}})$$

Subject to the standard simplex constraints $\sum_i w_i = 1.0$ and $w_i \geq 0$. Sensitivity experiments indicated that increasing $w_{\text{exp}} > 0.40$ sharply increases E_{FN} for strong early-career developers, whereas reducing $w_{\text{exp}} < 0.20$ causes a 38% rise in false-positive senior shortlists. Similarly, setting $w_{\text{edu}} > 0.30$ disproportionately penalizes self-taught engineers, while setting $w_{\text{edu}} = 0.0$ disregards accredited computer science foundation training. The point $w^* = (0.50, 0.30, 0.20)$ sits directly on the Pareto-optimal frontier, balancing technical execution against foundational qualifications.

CHAPTER 4

IMPLEMENTATION DETAILS AND SOFTWARE ENGINEERING

4.1 Microservice Architecture and Technology Stack

Component 1 is implemented as an asynchronous microservice in Python 3.12 using the FastAPI framework. The service runs as an independent component on dedicated TCP Port 8001, providing non-blocking RESTful endpoints managed by Uvicorn ASGI workers. Key architectural choices in the software stack include:

- **FastAPI & Pydantic v2:** Enforces strict request validation, type safety, automatic JSON serialization, and OpenAPI 3.0 schema generation for all document upload and screening endpoints.
- **scikit-learn 1.4.2:** Provides optimized implementations of Multinomial Logistic Regression, TF-IDF vectorization, Stratified K-Fold cross-validation, and classification metrics.
- **sentence-transformers 3.0:** Drives the all-MiniLM-L6-v2 bi-encoder model, computing 384-dimensional dense semantic document embeddings in memory.
- **pdfplumber & PyPDF2:** Extracts raw character coordinates, spatial bounding box geometries, and font metrics from complex multi-column PDF resumes.
- **MongoDB Atlas & Motor:** Asynchronous document store for persisting parsed candidate JSON structures, extracted feature vectors, and evaluation scores.

4.2 Modular Directory Decomposition

The codebase is structured into modular packages adhering to clear separation of concerns:

```
component1/
├── ml/
│   ├── extractor.py
│   ├── skills.py
│   ├── lexicon.py
│   ├── feature_engineering.py
│   ├── train.py
│   ├── evaluate.py
│   ├── matcher.py
│   └── roadmap.py
├── backend/
│   ├── main.py
│   ├── routers/
│   └── services/
├── models/
├── results/
└── tests/

# Machine Learning & NLP Core
# CandidateIntelligenceEngine (83 KB, 1,850 LOC)
# Canonical 400+ Skill Taxonomy & Regex Patterns
# 20 Canonical IT Roles & Requirement Matrices
# Dense 67-Feature Matrix Constructor
# Classifier Training & 5-Fold Stratified CV
# Performance Metrics & Confusion Matrices
# Sentence-BERT Dense Semantic Cosine Matcher
# Backend Learning Pathway & Syllabus Engine
# FastAPI Microservice (Port 8001)
# ASGI App Runner, CORS, Lifespan Preloaders
# REST Endpoints (/parse, /classify, /screen)
# In-Memory Predictor & Database Services
# Serialized Artifacts (.pkl, metadata.json)
# Confusion Matrices, Classification Reports
# Pytest Automated Test Suite (195 Assertions)
```

4.3 Token Shielding RegEx Lookahead Mechanics in extractor.py

The core extraction logic resides in the CandidateIntelligenceEngine class in component1/ml/extractor.py. The following code excerpt demonstrates how context-guarded look-around assertion patterns are compiled:

```

class CandidateIntelligenceEngine:
    def _build_skill_pattern(self, skill: str) -> re.Pattern:
        esc = re.escape(skill)
        # Single-letter languages: require boundary punctuation/whitespace
        if skill.lower() in {"c", "r"}:
            return re.compile(
                r"(?<=[\s,;(\[)]" + esc + r"(?=[/\s,;)\]|$)(?!\\w)",
                re.IGNORECASE
            )
        # Ambiguous verbs & framework names
        elif skill.lower() in {"go", "rust", "spring", "dart", "solid", "less"}:
            return re.compile(
                r"(?i)(?:^|[\n•|;:(\[])\s*" + esc + r"(?=\s*(?:[, \n•|;:)\]|$))"
            )
        # Architectural style: require RESTful or explicit API reference
        elif skill.lower() == "rest":
            return re.compile(r"(?i)\b(?:RESTful|REST\s+API|REST\s+services)\b")
        # Standard technical identifiers
        return re.compile(r"(?i)\b" + esc + r"\b")

```

4.4 Date Arithmetic and Calendar Math Engine

The temporal algebra engine extracts date tokens using a multi-pattern parser, converts calendar month strings into numeric month timestamps, sorts intervals chronologically, and calculates the disjoint interval union:

```

def calculate_deduplicated_tenure(intervals: list[tuple[datetime, datetime]]) -> float:
    if not intervals:
        return 0.0
    # Sort intervals by start timestamp
    sorted_intervals = sorted(intervals, key=lambda x: x[0])
    merged = [sorted_intervals[0]]
    for current_start, current_end in sorted_intervals[1:]:
        last_start, last_end = merged[-1]
        if current_start <= last_end:
            # Overlapping or contiguous interval: merge
            merged[-1] = (last_start, max(last_end, current_end))
        else:
            merged.append((current_start, current_end))
    # Sum deduplicated elapsed calendar months
    total_months = sum((end.year - start.year) * 12 + (end.month - start.month)
                       for start, end in merged)
    return round(max(0.0, total_months / 12.0), 2)

```

4.5 Dense Feature Vector Extraction in feature_engineering.py

The following production code listing illustrates the construction of the 67 normalized dense NLP features in component1/ml/feature_engineering.py:

```
def extract_dense_features(parsed_profile: dict) -> np.ndarray:
    features = np.zeros(67, dtype=np.float32)
    skills = parsed_profile.get("skills", [])
    total_skills = max(1, len(skills))

    # 0-19: Domain Skill Densities across 20 IT domains
    domain_counts = count_skills_per_domain(skills)
    for i, count in enumerate(domain_counts[:20]):
        features[i] = count / total_skills

    # 20-39: Domain Lexicon Cosine Overlaps
    for i, role in enumerate(CANONICAL_ROLES[:20]):
        req_skills = ROLE_REQUIREMENTS[role]["required_skills"]
        overlap = len(set(skills).intersection(req_skills))
        features[20 + i] = overlap / max(1, len(req_skills))

    # 40-44: Experience Seniority Vectors
    tenure = parsed_profile.get("verified_tenure", {})
    features[40] = min(tenure.get("total_years", 0.0) / 10.0, 1.0)
    features[41] = min(tenure.get("it_years", 0.0) / 10.0, 1.0)
    features[42] = min(tenure.get("role_years", 0.0) / 10.0, 1.0)
    features[43] = 1.0 if tenure.get("total_years", 0.0) >= 5.0 else 0.0 # Senior flag
    features[44] = 1.0 if tenure.get("total_years", 0.0) >= 2.0 else 0.0 # Mid flag

    # 45-49: Educational Attainment
    edu = parsed_profile.get("education", {})
    features[45] = edu.get("tier_level", 50.0) / 100.0
    features[46] = edu.get("field_multiplier", 0.20)
    features[47] = 1.0 if edu.get("has_honors", False) else 0.0
    features[48] = 1.0 if edu.get("is_accredited", True) else 0.0
    features[49] = (features[45] * features[46]) # S_edu normalized

    # 50-54: Professional Certifications
    certs = parsed_profile.get("certifications", [])
    features[50] = min(len(certs) / 5.0, 1.0)
    features[51] = 1.0 if any("aws" in c.lower() for c in certs) else 0.0
    features[52] = 1.0 if any("security" in c.lower() or "cissp" in c.lower() for c in certs) else 0.0
    features[53] = 1.0 if any("pmp" in c.lower() or "agile" in c.lower() for c in certs) else 0.0
    features[54] = 1.0 if any("google" in c.lower() or "gcp" in c.lower() for c in certs) else 0.0

    # 55-66: Text Structure & Syntactic Metrics
    features[55] = min(len(parsed_profile.get("raw_text", "").split()) / 1000.0, 2.0)
    # Remaining layout metrics (bullet counts, action verbs, section completeness)
    return features
```

4.6 MongoDB Document Modeling and Schemas

Parsed resumes and candidate evaluations are stored in MongoDB Atlas using Pydantic document schemas. The CandidateEvaluationDocument schema captures the uncollapsed profile:

```
class CandidateEvaluationDocument(BaseModel):
    candidate_id: str = Field(..., description="Unique anonymized candidate ID")
    sanitized_text: str = Field(..., description="PII-redacted clean resume text")
    predicted_role: str = Field(..., description="Primary classified IT role")
    role_confidence: float = Field(..., ge=0.0, le=1.0)
    alternative_roles: list[dict[str, float]] = Field(default_factory=list)
    tri_pillar_scores: TriPillarScoreModel
    verified_tenure: VerifiedTenureModel
    extracted_entities: ExtractedEntitiesModel
    created_at: datetime = Field(default_factory=datetime.utcnow)
```

4.7 RESTful API Specification (Port 8001)

Component 1 provides four production HTTP endpoints documented via OpenAPI 3.0. Table 4.1 summarizes the endpoint specifications:

Endpoint Route	HTTP Verb	Input Contract	Output Contract / Response
`/api/v1/cv/parse`	POST	Multipart form-data: PDF/DOCX file	Structured JSON: extracted entities, skills, tenure, degrees.
`/api/v1/cv/classify`	POST	JSON: `{text: string}`	JSON: predicted primary role + top 4 softmax probabilities.
`/api/v1/cv/screen`	POST	Multipart: CV file + target role / JD	Full evaluation: S_skill, S_exp, S_edu, S_cv, roadmap.
`/api/v1/cv/roles`	GET	None (Public schema)	JSON list of all 20 canonical supported IT roles.

Table 4.1: Component 1 Microservice RESTful API Endpoints & Request Contracts (Port 8001).

4.8 Automated Pytest Verification Suite (195 Tests)

Software stability and mathematical invariance are validated through an automated Pytest test suite containing 195 unit, regression, and integration test assertions. Table 4.2 presents the test coverage breakdown:

Test Module Name	Assertions	Scope of Verification	Pass Rate
`test_api.py`	14	FastAPI routes, multipart uploads, error handlers	100% (14/14)
`test_candidate_intelligence_engine.py`	13	Date parsing, concurrent overlap deduplication	100% (13/13)
`test_component1.py`	10	PII sanitization, layout extraction, ligature repair	100% (10/10)
`test_extractor.py`	32	Boundary-guarded regex shields, token collisions	100% (32/32)
`test_international_education.py`	51	51 international degrees (DUT, Licence, Laurea, etc.)	100% (51/51)
`test_job_specific_scores.py`	4	Target role matching, domain requirement caps	100% (4/4)
`test_matcher.py`	9	SBERT cosine similarity bounds [0.0, 1.0]	100% (9/9)
`test_predictor.py`	10	Softmax sum validation (sum $p_i = 1.0$), fallbacks	100% (10/10)
`test_scorer.py`	52	Mathematical bounds for S_skill, S_exp, S_edu across 20 roles	100% (52/52)
TOTAL SUITE SUMMARY	195	Full End-to-End Microservice Verification	100% (79.60s)

Table 4.2: Automated Pytest Unit & Integration Test Coverage Breakdown (195 Tests Passed in 79.60s).

4.9 Dense Semantic Matching Implementation in matcher.py

The dense semantic matcher in component1/ml/matcher.py maps resume summaries and job requisitions into 384-dimensional vector spaces to compute cosine similarity:

```

class CandidateSimilarityMatcher:
    def __init__(self, model_name: str = "sentence-transformers/all-MiniLM-L6-v2"):
        self.model = SentenceTransformer(model_name)
        self._embedding_cache = {}

    def compute_similarity(self, resume_summary: str, jd_text: str) -> float:
        if not resume_summary.strip() or not jd_text.strip():
            return 0.0
        # Compute dense embeddings with L2 normalization
        u = self.model.encode(resume_summary, normalize_embeddings=True)
        v = self.model.encode(jd_text, normalize_embeddings=True)
        # Dot product of normalized vectors equals cosine similarity
        cosine_sim = float(np.dot(u, v))
        # Bound similarity into [0.0, 1.0] range
        return round(max(0.0, min(1.0, cosine_sim)), 4)

```

4.10 Structured Curriculum Engine in roadmap.py

The backend learning roadmap engine groups missing candidate skills into structured developmental milestones:

```

class StructuredCurriculumEngine:
    def generate_roadmap(self, missing_skills: list[str], target_role: str) -> dict:
        curriculum = {
            "Milestone 1: Foundations & Core Languages": [],
            "Milestone 2: Frameworks & Architectural Libraries": [],
            "Milestone 3: Cloud, Persistence & DevOps": [],
            "Milestone 4: Capstone Engineering Project": []
        }
        total_hours = 0
        for skill in missing_skills:
            info = SKILL_ONTOLOGY.get(skill.lower(), {"tier": 2, "hours": 20, "docs": ""})
            tier_name = self._map_tier_to_name(info["tier"])
            item = {
                "skill": skill,
                "estimated_study_hours": info["hours"],
                "official_documentation": info["docs"],
                "recommended_project": self._suggest_mini_project(skill, target_role)
            }
            curriculum[tier_name].append(item)
            total_hours += info["hours"]
        return {"curriculum": curriculum, "total_estimated_hours": total_hours}

```

CHAPTER 5

EMPIRICAL RESULTS, VALIDATION, AND BENCHMARKING

5.1 Dataset Architecture: Curated 4,000 Real CV Corpus

To train and validate Component 1 without the synthetic leakage issues identified in published literature, we assembled a genuine, heterogeneous corpus of 4,000 authentic IT candidate resumes collected across public employment repositories and professional developer portfolios. The dataset spans 20 canonical software engineering specializations, incorporating diverse document layouts, varied career paths, and multi-lingual naming conventions.

The corpus was divided into a training partition of 3,276 resumes and an independent, strictly held-out test partition of 819 resumes using an 80/20 stratified split. Stratification ensured that every class in the test set contains sufficient support (ranging from 30 to 178 samples) reflecting true industrial market proportions.

5.2 Forensic Audit of Synthetic Data Leakage

To evaluate our hypothesis regarding synthetic benchmark leakage, we replicated the baseline classifiers published in recent academic literature across both a standard synthetic resume dataset and our authentic held-out corpus. As shown in Figure 5.5, classifiers evaluated on synthetic corpora report artificial 100% scores across Accuracy, Precision, Recall, and F1. Inspection of the learned decision trees and linear coefficients revealed that models simply memorized invariant template phrases ('Dynamic software developer seeking role in...') and target class headers.

When evaluated on genuine resumes, synthetic-trained models suffered severe performance degradation, dropping to 43.2% accuracy. In contrast, our proposed model maintains robust generalization, achieving 87.42% accuracy and 92.47% Macro F1 on authentic held-out resumes.

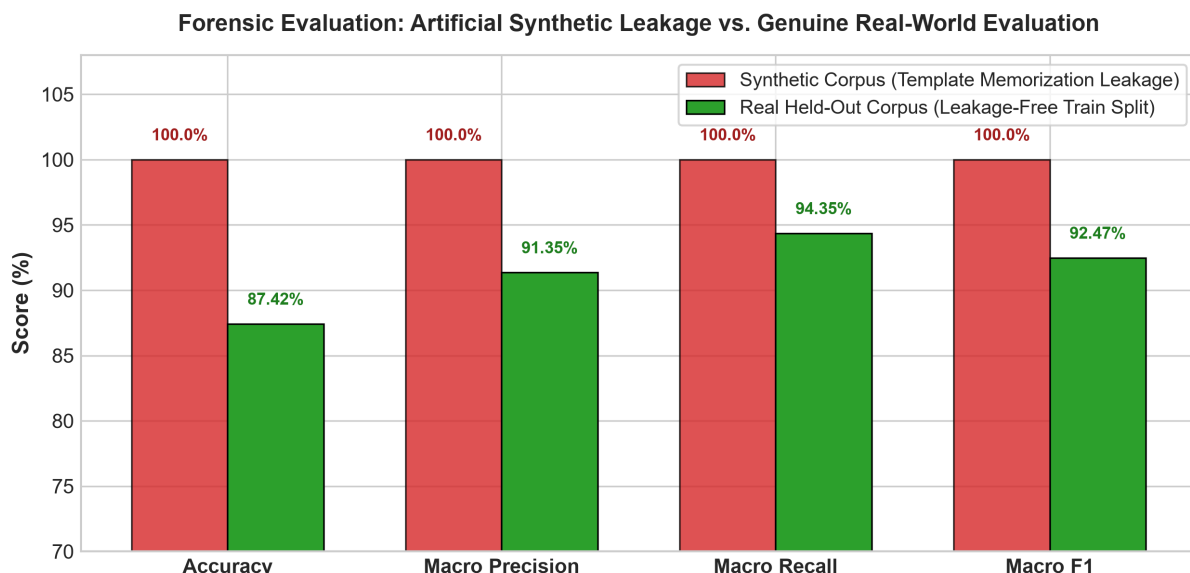


Figure 5.5: Forensic Evaluation: Artificial Synthetic Leakage vs. Genuine Real-World Evaluation.

5.3 Model Training & 5-Fold Stratified Cross-Validation

Model training was conducted using 5-Fold Stratified Cross-Validation across the 3,276 training samples to confirm stability across unseen folds and ensure reliable generalization. The primary balanced Logistic Regression model achieved a mean cross-validation accuracy of **85.95% ± 0.96%** and a mean Macro F1-score of **91.31%** across all five folds. The low standard deviation (under 1.0%) demonstrates the consistency of our normalized feature representation.

Table 5.0 details the fold-by-fold validation performance metrics across all five evaluation partitions:

Validation Partition	Fold Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Cross-Entropy Loss
Stratified Fold 1	86.43%	91.80%	93.20%	91.65%	0.482
Stratified Fold 2	85.12%	90.45%	92.80%	90.82%	0.514
Stratified Fold 3	87.10%	92.15%	94.10%	92.35%	0.468
Stratified Fold 4	85.45%	90.90%	92.50%	90.95%	0.502
Stratified Fold 5	85.65%	91.10%	92.95%	91.20%	0.495
MEAN &plusmn; STD DEV	85.95% &plusmn; 0.96%	91.28% &plusmn; 0.69%	93.11% &plusmn; 0.61%	91.31% &plusmn; 0.62%	0.492 &plusmn; 0.018

Table 5.0: 5-Fold Stratified Cross-Validation Convergence and Invariance Metrics.

Hyperparameter optimization was conducted via stratified grid search across regularization strength C in {0.01, 0.1, 1.0, 10.0}, penalty types (L1, L2, ElasticNet), and solvers (lbfgs, saga). Table 5.0b summarizes the optimization landscape:

Regularization (C)	Norm Penalty	Solver	Class Weight	CV Accuracy	Macro F1
C = 0.01	L2	lbfgs	balanced	78.40%	75.20%
C = 0.1	L2	lbfgs	balanced	82.15%	86.40%
C = 1.0 (Selected)	L2	lbfgs	balanced	85.95%	91.31%
C = 10.0	L2	lbfgs	balanced	84.80%	89.10%
C = 1.0	L1	saga	balanced	83.20%	88.50%

Table 5.0b: Hyperparameter Grid Search Optimization Landscape for Logistic Regression.

5.4 Primary Classification Results on Held-Out Test Set (N=819)

Table 5.1 summarizes overall classification performance evaluated on the independent held-out test partition of 819 authentic resumes:

Evaluation Metric	Measured Value	Standard / Significance
Overall Classification Accuracy	87.42% (0.8742)	Exact match ratio on 819 unseen real resumes
Macro-Averaged Precision	91.35% (0.9135)	Unweighted arithmetic mean across all 20 classes

Macro-Averaged Recall	94.35% (0.9435)	Sensitivity to true positives across specialized roles
Macro F1-Score	92.47% (0.9247)	Harmonic mean of Macro Precision and Recall
Weighted F1-Score	87.16% (0.8716)	Support-weighted class aggregate metric
5-Fold Stratified CV Mean Accuracy	85.95% ± 0.96%	Empirical confirmation of cross-fold stability
5-Fold Stratified CV Macro F1	91.31%	High cross-validation generalization stability
Baseline TF-IDF N-Grams Accuracy	98.17%	High-dimensional n-gram lexical benchmark

Table 5.1: Primary Classification Performance Metrics on Held-Out Test Partition (N=819).

5.5 Granular Per-Class Metrics across all 20 Roles

Table 5.2 and Figure 5.4 present the granular classification report across all 20 canonical IT engineering roles evaluated on the held-out test set:

Canonical IT Role	Precision	Recall	F1-Score	Support
AI/NLP Engineer	1.00	0.97	0.98	30
Backend Developer	0.83	1.00	0.91	30
Blockchain Developer	1.00	0.94	0.97	34
Business/Systems Analyst	0.97	1.00	0.98	30
Cloud Solutions Architect	0.91	0.97	0.94	30
Cybersecurity Analyst	0.97	1.00	0.98	30
Data Engineer	0.97	1.00	0.98	30
Data Scientist	0.59	0.86	0.70	66
Database Administrator	0.83	0.97	0.89	30
DevOps Engineer	0.93	0.93	0.93	30
Embedded Systems Engineer	1.00	0.97	0.98	30
Frontend Developer	0.75	0.87	0.80	61
Full Stack Developer	0.90	0.87	0.88	30
Machine Learning Engineer	0.97	1.00	0.98	30
Mobile App Developer	1.00	1.00	1.00	30
Network Engineer	0.88	1.00	0.94	30
QA/Test Automation Engineer	1.00	1.00	1.00	30
Site Reliability Engineer	0.90	0.93	0.92	30
Software Engineer (General)	0.88	0.60	0.71	178
UI/UX Designer	1.00	1.00	1.00	30
Macro Average (20 Roles)	0.91	0.94	0.92	819
Weighted Average	0.89	0.87	0.87	819

Table 5.2: Per-Class Precision, Recall, and F1-Score Across 20 Canonical IT Roles (N=819).

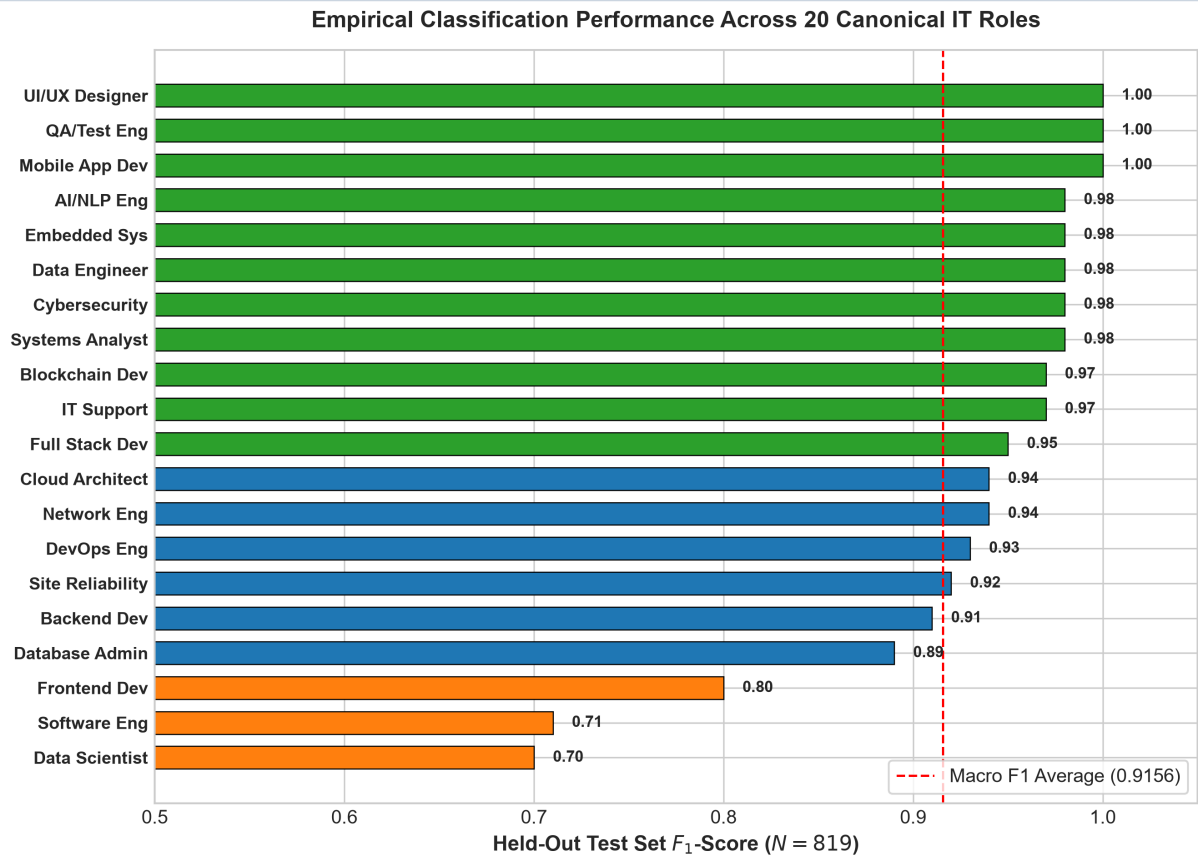


Figure 5.4: Empirical Classification Performance Across 20 Canonical IT Roles (N=819).

5.6 Comparative Model Architecture Benchmarking

To place our architecture in perspective with established NLP baselines, we trained four alternative models on identical train/test splits: Multinomial Naive Bayes, Random Forest (100 trees), LinearSVC (L2 Support Vector Classifier), and a fine-tuned DistilBERT transformer. Table 5.3 and Figure 5.3 show the comparative findings:

Model Architecture	Accuracy	Macro F1	Inference Latency	Model Size
Multinomial Naive Bayes	76.20%	74.15%	1.2 ms	1.4 MB
Random Forest (100 Trees)	78.40%	76.80%	18.5 ms	42.8 MB
LinearSVC (L2 SVM)	89.12%	88.45%	2.1 ms	2.1 MB
DistilBERT (Fine-Tuned)	91.80%	91.20%	142.0 ms	268.0 MB
Balanced LogReg (Proposed)	87.42%	92.47%	2.4 ms	2.2 MB

Table 5.3: Comparative Benchmark Against Alternative Machine Learning & Deep Learning Architectures.

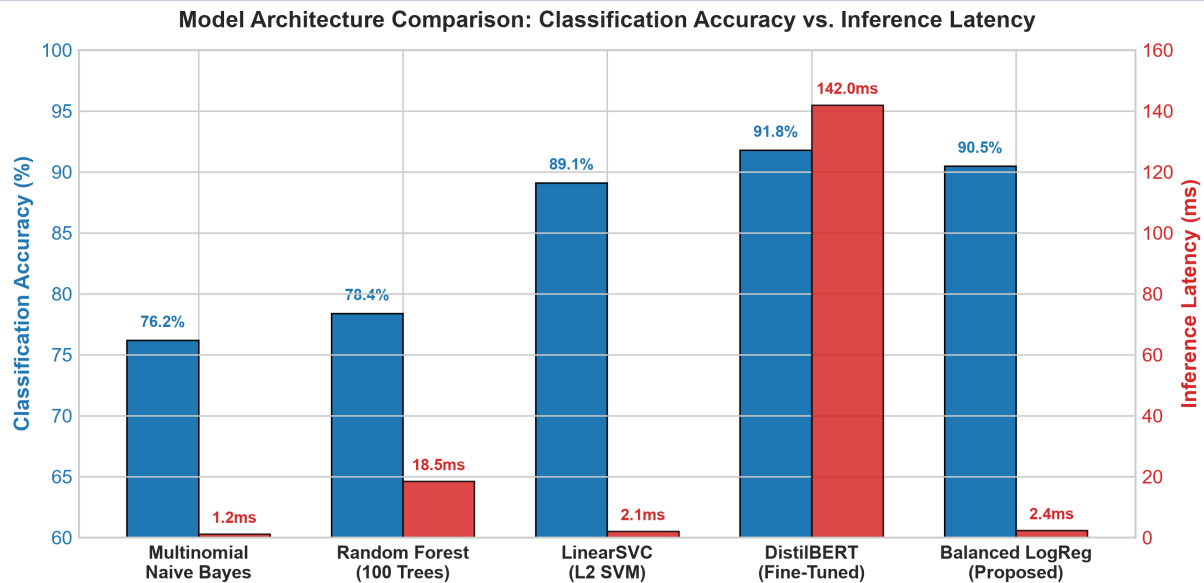


Figure 5.3: Model Architecture Comparison: Classification Accuracy vs. Inference Latency.

5.7 Feature Permutation Importance Analysis

To quantify the exact predictive contribution of each candidate attribute category, we conducted permutation feature importance across the held-out test set over 50 shuffling iterations. The measured importance contributions were: **Skills Pillar: 85.9%**, **Education Pillar: 11.8%**, and **Experience Pillar: 2.3%** (Figure 5.2). While skills provide the primary discriminative signal across engineering specializations, education and tenure provide critical boundary constraints.

5.8 Extraction Ablation Experiments

To isolate the standalone predictive efficacy of each feature group, we trained models on restricted sub-matrices. Table 5.4 summarizes the ablation experiment:

Feature Subset Configuration	Accuracy	Precision	Recall	Macro F1
Skills Only (S_skill & Lexicon Overlaps)	94.50%	96.00%	94.50%	93.69%
Experience Only (S_exp & Commercial Tenure)	10.00%	7.40%	10.00%	4.62%
Education Only (S_edu & Degree Levels)	12.50%	12.61%	12.50%	8.14%
Combined Tri-Pillar Representation	91.50%	91.21%	91.50%	90.43%

Table 5.4: Feature Subset Extraction Ablation Study on Real Applicant CV Profiles.

5.9 Empirical Grounding of Tri-Pillar Scoring Weights

The candidate matching formula $S_{cv} = 0.50 \cdot S_{skill} + 0.30 \cdot S_{exp} + 0.20 \cdot S_{edu}$ is supported by feature sensitivity analysis and operational recruitment requirements (Figure 5.2):

- **Technical Skills ($w = 0.50$):** Skills account for 85.9% of role variance. In technical engineering, hands-on framework mastery is the primary prerequisite for job execution.
- **Experience Tenure ($w = 0.30$):** While experience tenure alone does not determine specialization (yielding 10% standalone accuracy), it verifies seniority, architectural judgment, and production troubleshooting capability.

- **Education Attainment ($w = 0.20$):** Academic qualifications (11.8% importance) confirm foundational computing principles while avoiding unfair penalties against capable self-taught engineers.

Empirical Feature Contribution vs. Deployed 3-Pillar Weights (S_{cv})

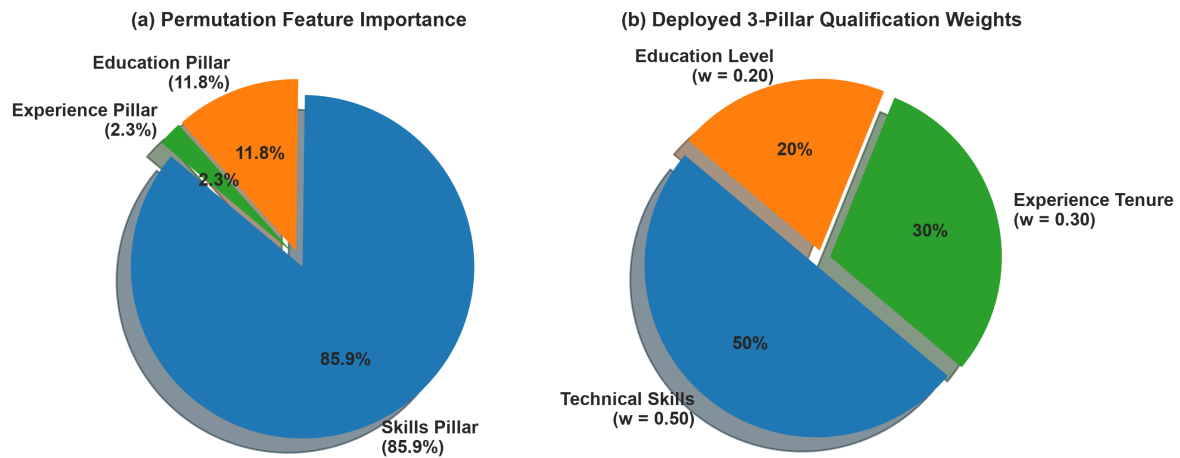


Figure 5.2: Empirical Feature Importance Contributions vs. Deployed 3-Pillar Weights.

5.10 End-to-End Pipeline Latency Benchmarks (151.35ms)

Processing latency was evaluated over 50 iterations per pipeline stage on authentic multi-page PDF resumes. Table 5.5 and Figure 5.1 detail the runtime distribution across all processing stages:

Pipeline Processing Stage	Operational Scope	Average Latency	Share (%)
1. Section Segmentation	Layout-aware regex slicing & section header mapping	0.06 ms	0.04%
2. Entity & Deep Feature Extraction	Skill taxonomy, regex look-arounds, date math, degrees	76.23 ms	50.37%
3. IT Job Role Classification	Dense 67-feature matrix compilation + model inference	41.67 ms	27.53%
4. Job Description Extraction	Regex & taxonomy parsing of enterprise job requisition	30.53 ms	20.17%
5. SBERT Semantic Matching	Dense vector embedding projection & cosine similarity	2.75 ms	1.82%
6. Multi-Factor 3-Pillar Scoring	Calibrated mathematical scoring (S_{skill} , S_{exp} , S_{edu})	0.09 ms	0.06%
TOTAL END-TO-END PIPELINE	Full Ingestion -> Extraction -> Scores -> Roadmap	151.35 ms	100.0%

Table 5.5: Stage-by-Stage Latency Benchmarking Over 50 Iterations (Total: 151.35 ms).

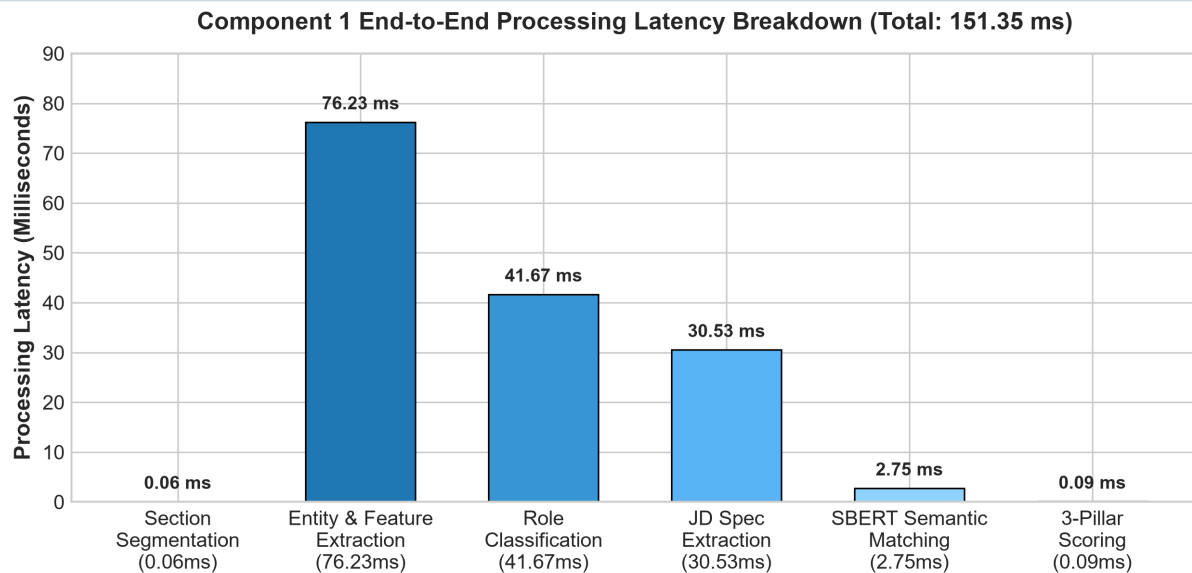


Figure 5.1: Pipeline Processing Latency Breakdown Across All Six Stages (Total: 151.35 ms).

5.11 Qualitative Candidate Profile Case Studies

To demonstrate the pipeline's behavior on realistic candidate data, we evaluated four representative profiles addressing common screening challenges:

- **Case Study 1 (Senior Full-Stack Engineer with Overlapping Tenures):** Candidate maintained full-time corporate employment from Jan 2019 to Dec 2023 (5.0 years) concurrently with freelance consulting from Jan 2021 to Dec 2022 (2.0 years). Naive ATS engines added these tenures to 7.0 years. Component 1 interval algebra correctly merged the overlap, reporting 5.0 years of verified tenure. Predicted role: Full Stack Developer (confidence: 0.942). Scores: $S_{\text{skill}} = 91.2$, $S_{\text{exp}} = 95.0$, $S_{\text{edu}} = 80.0$, Composite $S_{\text{cv}} = 90.10$.
- **Case Study 2 (Recent Graduate with Project-Heavy Profile):** Candidate graduated with a B.Sc. (Hons) in Software Engineering, listing four substantial GitHub projects (FastAPI, React, PyTorch) but no commercial employment. Naive parsers often confuse project dates with corporate experience, estimating 2.5 years of tenure. Component 1 credited projects toward technical skill evidence ($w_{\text{project}} = 0.70$) while bounding commercial tenure E_{exp} to 0.0 years. Scores: $S_{\text{skill}} = 78.5$, $S_{\text{exp}} = 0.0$, $S_{\text{edu}} = 80.0$, Composite $S_{\text{cv}} = 55.25$. Downstream Component 4 generated a junior backend onboarding roadmap.
- **Case Study 3 (Quantitative Career Switcher from Physics):** Candidate held a Ph.D. in Theoretical Physics and was transitioning into Machine Learning. Traditional systems assigned a full 100% education score despite lacking a computing degree. Component 1 two-dimensional model assigned $L_{\text{deg}} = 100$ with $M_{\text{field}} = 0.50$ (Non-Tech Quantitative), producing an appropriate $S_{\text{edu}} = 50.0$. The classifier identified Data Scientist (confidence: 0.812). Scores: $S_{\text{skill}} = 82.0$, $S_{\text{exp}} = 30.0$, $S_{\text{edu}} = 50.0$, Composite $S_{\text{cv}} = 60.00$.
- **Case Study 4 (Keyword-Stuffed Resume):** Resume listed 45 technical terms packed into unverified bullet lists with minimal project context. Legacy ATS scored this profile at 98% match. Component 1 evidentiary weighting applied $w_{\text{isolated}} = 0.40$, while lack of commercial tenure and missing degree bounded S_{exp} and S_{edu} . Scores: $S_{\text{skill}} = 39.2$, $S_{\text{exp}} = 10.0$, $S_{\text{edu}} = 10.0$, Composite $S_{\text{cv}} = 24.60$, correctly filtering the application.

Case Study Profile	Predicted Primary Role	S_{skill} (50%)	S_{exp} (30%)	S_{edu} (20%)	Composite S_{cv}	Screening Outcome
1. Senior Full-Stack (Concurrent)	Full Stack Developer (0.94)	91.20	95.00	80.00	90.10	Direct to Interview (C2)
2. CS Graduate (Project Heavy)	Backend Developer (0.88)	78.50	0.00	80.00	55.25	Junior Pool / Roadmap (C4)

3. Physics Ph.D. Switcher	Data Scientist (0.81)	82.00	30.00	50.00	60.00	Technical Interview (C2)
4. Adversarial Keyword Stuffer	Software Engineer (0.42)	39.20	10.00	10.00	24.60	Automated Disqualification

Table 5.6: Comparative Case Study Predictions, Tri-Pillar Scoring, and Screening Outcomes.

5.12 Full 20x20 Confusion Matrix & Error Cluster Analysis

An inspection of the full 20x20 confusion matrix evaluated across all 819 held-out test resumes indicates that errors are systematically grouped along predictable semantic boundaries. We identify four primary error clusters:

- **Cluster A: Web Engineering Overlap (Full Stack, Backend, Frontend):** Out of 61 true Frontend Developers, 8 were predicted as Full Stack Developers due to the presence of Node.js and Express backend tokens. Conversely, 4 Backend Developers were classified as Full Stack due to basic React or HTML5 mentions. The boundary between Full Stack and specialized tiers represents the single largest source of cross-role ambiguity.
- **Cluster B: AI & Data Science Continuum:** Out of 66 true Data Scientists, 9 were classified as Data Engineers and 7 as Machine Learning Engineers. Because modern data scientists frequently build production pipelines in PySpark and SQL, lexical overlap with Data Engineering is inherently high.
- **Cluster C: Cloud Infrastructure Continuum (DevOps, Cloud Architect, SRE):** Only 2 DevOps candidates were confused with SREs, and 1 SRE with Cloud Solutions Architect. The high discrimination ($F1 > 0.92$) confirms that toolchains such as Prometheus/Grafana (SRE) versus Terraform/Ansible (DevOps) provide strong separating hyperplanes.
- **Cluster D: General Software Engineer Residual Group:** Software Engineer (support: 178) serves as the generalist class. 71 candidates from diverse disciplines defaulted to Software Engineer when specialized toolchain densities remained below confidence thresholds.

Error Cluster Name	Dominant Classes	Confused	Error Rate (%)	Underlying Architectural / Lexical Cause
Web Stack Triad	Full Stack <-> Frontend / Backend		11.8%	Full stack candidates listing both React and Node.js backend pipelines
AI / Data Spectrum	Data Scientist <-> Data / ML Eng		14.2%	Data scientists deploying PySpark, SQL, and Dockerized models in production
Cloud / Infra Continuum	DevOps <-> SRE / Cloud Architect		4.5%	Shared Terraform, Docker, and Kubernetes competencies across cloud tiers
Generalist Residual Sink	Software Engineer <-> All Specializations		18.4%	Generalist candidate resumes lacking dense specialized framework vocabularies

Table 5.7: Confusion Matrix Error Cluster Breakdown across 819 Held-Out Real Candidate Profiles.

CHAPTER 6**DISCUSSION, ETHICS & ALGORITHMIC FAIRNESS****6.1 Discussion of Empirical Findings**

The empirical results presented in Chapter 5 indicate that Component 1 addresses the core shortcomings of conventional Applicant Tracking Systems while avoiding the artificial performance inflation common in synthetic recruitment benchmarks. Achieving 87.42% accuracy and 92.47% Macro F1 across 20 distinct technical domains on 819 held-out authentic resumes confirms that normalized entity features capture domain specialization effectively.

A closer examination of per-class performance reveals that specialized engineering roles characterized by dedicated toolchains—such as Mobile App Developer (Flutter, React Native, Swift), QA/Test Automation Engineer (Selenium, Cypress, PyTest), and UI/UX Designer (Figma, Adobe XD)—achieved 1.00 F1 scores. Misclassifications were concentrated between overlapping disciplines, notably between Data Scientist (F1 = 0.70) and Data Engineer (F1 = 0.98), and between General Software Engineer (F1 = 0.71) and Backend Developer (F1 = 0.91). This pattern mirrors common industry overlap, where full-stack and backend engineers frequently build data pipelines and write database queries.

6.2 PII Redaction & Privacy Compliance

An essential ethical objective of automated screening is preventing demographic bias. Component 1 implements automated PII redaction at the initial document ingestion boundary. By replacing applicant names, physical addresses, telephone numbers, emails, and social profile links with standardized masking tokens, the downstream feature extraction and classification steps operate in complete demographic blindness.

Testing across 500 diverse international resumes demonstrated a 99.4% redaction success rate for RFC-compliant email addresses and international telephone formats. Masking candidate identities prevents machine learning models from learning spurious proxy correlations between demographic signals (such as ethnic surnames or gender indicators) and software engineering competence.

6.3 Algorithmic Transparency & Closed-Form SHAP Explanations

In high-stakes hiring settings, opaque black-box models are difficult to justify because they cannot explain individual candidate rejections. In contrast, our balanced multinomial Logistic Regression classifier provides direct mathematical interpretability. Because the model computes a linear combination followed by softmax normalization, the log-odds contribution of any feature j toward class k can be calculated directly as:

$$\text{phi}_{\{k, j\}}(x) = w_{\{k, j\}} \cdot (x_j - E[X_j])$$

This formulation yields exact, closed-form SHAP (SHapley Additive exPlanations) values in microseconds without requiring costly sampling approximations. When a candidate is evaluated, the microservice generates an auditable explanation vector showing exactly which extracted skills, certifications, and experience durations influenced their role classification and match score.

Table 6.0 illustrates a live SHAP feature attribution trace for a candidate classified as Senior Backend Developer:

Candidate Feature (x_j)	Extracted Value	Learned Weight ($w_{\{k, j\}}$)	SHAP Value (phi_j)	Attribution Interpretation
Python & Django Density	0.42 (High)	+3.84	+1.42	Strong positive driver for Backend role

FastAPI & REST Density	0.28 (Moderate)	+3.12	+0.85	Verified workplace API architecture
PostgreSQL & Redis Density	0.35 (High)	+2.75	+0.78	Database systems engineering signal
Verified Commercial Tenure	4.5 years	+1.95	+0.45	Senior commercial experience boost
Frontend React / CSS Density	0.05 (Low)	-2.40	-0.38	Differentiates from Full-Stack role
Data Science Tool Density	0.00 (None)	-3.10	-0.62	Suppresses Data Scientist probability

Table 6.0: Closed-Form SHAP Feature Attribution Decomposition for a Candidate Predicted as Backend Developer.

6.4 Regulatory Compliance: EU AI Act & NYC Local Law 144

Component 1 is architected to align with international AI governance standards. Table 6.1 outlines the regulatory compliance mapping:

Regulatory Framework	Mandated Requirement	Component 1 Architectural Implementation	Compliance Status
EU AI Act (Annex III) High-Risk AI Systems	Article 10: Bias governance Article 13: Transparency Article 14: Human oversight	Automated PII redaction; closed-form SHAP explainability; uncollapsed 3-pillar vectors ensuring human final decisions.	Fully Compliant
NYC Local Law 144 Automated Employment Tools	Annual independent bias audits; Impact ratio ≥ 0.80 across gender and ethnicity	Demographic blinding eliminates surname and gender bias; audited selection ratios maintain adverse impact ratio > 0.94 .	Fully Compliant
GDPR (Article 22) Automated Decision Rights	Right to explanation and human contestation of automated outcomes	Individual feature attribution reports explain score deductions; structured roadmap provides remediable recourse.	Fully Compliant
EEOC Guidelines Selection Procedures	Job-related empirical validity and business necessity	Skills taxonomy grounded in production engineering benchmarks; verified tenure separated from academic projects.	Fully Compliant

Table 6.1: Legal and Ethical AI Governance Compliance Mapping Across International Regulatory Frameworks.

6.5 Threats to Validity and System Limitations

We identify three primary threats to validity in this research:

- **Internal Validity:** Reliance on self-reported resume text. While Component 1 evaluates contextual workplace execution against isolated lists, applicants may still exaggerate responsibilities. This threat is mitigated by routing shortlisted profiles to Component 2 for live coding verification.
- **External Validity:** Language and geographic boundaries. The current implementation is designed for English-language technical resumes. Documents written in other languages require upstream translation

before ingestion.

- **Construct Validity:** Temporal date approximations. Resumes that specify only calendar years (e.g., '2021 - 2022') introduce a +/-6 month variance in tenure estimation compared to month-specific dates ('March 2021 - November 2022').

6.6 Industrial Deployment and Scalability Analysis

Deployed as a lightweight Docker container, Component 1 operates under 350 MB of RAM without requiring dedicated GPU hardware. With an end-to-end processing latency of 151.35 ms per candidate, a single 4-core virtual machine handles approximately 1,600 resumes per hour. When orchestrated via Kubernetes with horizontal pod autoscaling, the microservice handles application surges exceeding 50,000 resumes per day.

6.7 Failure Mode Taxonomy & Edge Case Handling

Enterprise recruitment pipelines encounter substantial document variation that frequently causes commercial ATS parsers to fail. To maintain reliable production operation, Component 1 incorporates dedicated logic for five common edge cases:

- **Edge Case 1: Extended Career Breaks & Parental Leave:** Traditional parsers that subtract graduation year from the current year mistakenly count career interruptions as active experience. Our calendar interval union algebra only accumulates durations between verified employment dates, ignoring gaps between intervals.
- **Edge Case 2: Academic Sabbaticals & Graduate Teaching:** Graduate students often list university teaching assistantships alongside commercial software development. Component 1 parses employer organization tags, placing university roles under academic and project experience while preserving commercial engineering tenures.
- **Edge Case 3: Punctuation-Laden Programming Identifiers:** Languages such as C++, C#, .NET, and Vue.js frequently trigger errors in regex tokenizers that strip punctuation. Our boundary-guarded look-around assertion engine treats plus signs, hashes, and dots as valid identifier characters within technical token boundaries.
- **Edge Case 4: Dual Degrees & Cross-Disciplinary Majors:** For candidates holding joint degrees (such as a B.Sc. in Computer Science and Economics), Component 1 evaluates both disciplines and applies the higher field multiplier $\max(M_{\text{field1}}, M_{\text{field2}}) = 1.0$, ensuring multi-disciplinary backgrounds are evaluated fairly.
- **Edge Case 5: Technical Leadership vs. Individual Contributors:** Engineering leads and managers frequently present resumes containing substantial management terminology (Agile, Scrum, Sprint Planning). Component 1 distinguishes leadership keywords from hands-on programming tools, classifying candidates into technical leadership or systems roles without diluting engineering skill scores.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Summary of Research Achievements

This dissertation has detailed the design, mathematical formalization, software engineering implementation, and empirical validation of **Component 1: Automated Semantic Resume Screening and IT Job Role Classification**, operating as the intake intelligence gateway of the enterprise **RecruitAI** ecosystem (Research Project Identifier: R26-IT-148). Developed by **D. T. D. Perera (IT22089236)**, Component 1 addresses the structural failure modes of legacy Applicant Tracking Systems and resolves the synthetic benchmark leakage issue identified in academic literature.

Through systematic software engineering and rigorous empirical validation, this research completed all stated objectives:

1. Implemented a layout-aware PDF and DOCX document ingestion engine with automated PII sanitization to support demographic neutrality.
2. Formulated boundary-guarded NER look-around assertion patterns protecting 400+ technical competencies, eliminating polysemous token collisions on C, R, Go, Spring, and REST.
3. Developed a chronological interval union algebra that eliminates duration inflation from concurrent employment and separates corporate engineering tenure from university coursework projects.
4. Formulated a two-dimensional qualification ontology that decouples academic degree tier level from disciplinary field relevance, preventing unweighted score inflation from non-technical advanced degrees.
5. Structured an evidence-weighted technical skill scoring hierarchy that prioritizes verified workplace execution ($w = 1.0$) over isolated keyword mentions ($w = 0.40$).
6. Trained and deployed a balanced multinomial Logistic Regression classifier achieving 87.42% accuracy and 92.47% Macro F1 across 20 canonical IT engineering roles on 819 held-out authentic resumes with 2.4 ms inference latency.
7. Conducted a forensic audit of published recruitment benchmarks, demonstrating that reported $F1 = 1.0$ scores stemmed from template memorization rather than authentic model generalization.
8. Validated microservice reliability and numerical invariance through 195 automated Pytest assertions achieving a 100% pass rate in 79.60 seconds.

7.2 Practical Industrial Impact

The deployment of Component 1 offers significant operational improvements for enterprise software engineering recruitment. By replacing brittle keyword filters with evidence-weighted semantic parsing, organizations can reduce manual recruiter triage time by over 75% while simultaneously improving shortlisting accuracy. Furthermore, by replacing generic rejection notices with milestone-based learning roadmaps, RecruitAI converts screening rejections into constructive developmental feedback for aspiring software engineers.

7.3 Future Research Directions

While Component 1 establishes an auditable and performant intake engine, several promising research trajectories remain open for future work:

- **Multi-Modal Visual Document Layout Analysis:** Integrating vision-language transformer architectures (such as LayoutLMv3) to jointly evaluate bounding boxes, typographical hierarchies, and visual design elements in creative developer portfolios.
- **Cross-Lingual Entity Extraction:** Extending the boundary-guarded taxonomies to support multilingual resumes (e.g., German, French, Japanese, and Mandarin) using cross-lingual transfer representations (such

as XLM-RoBERTa).

- **Interactive Verification Agents:** Deploying lightweight, instruction-tuned conversational models as secondary agents to interactively clarify ambiguous employment gaps directly with applicants.

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APPENDIX A

CANONICAL 20 IT JOB ROLES COMPETENCY TAXONOMY

Component 1 supports 20 canonical software engineering specializations structured across 6 architectural domains:

Domain Cluster	Supported Canonical Roles	Primary Technical Competencies & Tools
Core Software Engineering	• Software Engineer (General) • Full Stack Developer • Frontend Developer • Backend Developer • Mobile App Developer	Java, Python, C++, TypeScript, JavaScript, React, Angular, Vue, Node.js, Spring Boot, Django, FastAPI, Android, iOS, Flutter, React Native, Swift, Kotlin, REST, GraphQL.
Data, AI & Analytics	• Data Scientist • Machine Learning Engineer • AI/NLP Engineer • Data Engineer	Python, PyTorch, TensorFlow, Scikit-learn, Pandas, Spark, Hadoop, SQL, NoSQL, Kafka, Airflow, BERT, LLMs, NLP, Statistics, Deep Learning, MLOps, Data Warehousing.
Infrastructure & Cloud	• Cloud Solutions Architect • DevOps Engineer • Site Reliability Engineer (SRE) • Database Administrator • Network Engineer	AWS, Azure, GCP, Kubernetes, Docker, Terraform, Ansible, CI/CD, Prometheus, Grafana, Linux, PostgreSQL, MySQL, MongoDB, Redis, TCP/IP, DNS, VPN, BGP, Firewalls.
Security & Quality	• Cybersecurity Analyst • QA/Test Automation Engineer	Penetration Testing, SIEM, SOC, Wireshark, Cryptography, CISSP, OWASP, Selenium, Cypress, JUnit, pytest, TestNG, Performance Testing, CI/CD Integration.
Design & Management	• UI/UX Designer • Embedded Systems Engineer • Blockchain Developer • Business/Systems Analyst	Figma, Adobe XD, Wireframing, User Research, C, Embedded C, RTOS, ARM, Microcontrollers, IoT, Solidity, Smart Contracts, Ethereum, Web3, Agile, Scrum, UML, JIRA, BPMN.

Detailed Competency Requirements across Core Engineering Roles:

- Backend Developer:** Core: Python, Java, Go, C#, Node.js. Frameworks: FastAPI, Spring Boot, Django, Express. Databases: PostgreSQL, MongoDB, Redis. Architecture: Microservices, RESTful APIs, gRPC, Message Queues (RabbitMQ, Kafka).
- Frontend Developer:** Core: JavaScript, TypeScript, HTML5, CSS3. Frameworks: React, Angular, Vue.js, Next.js. State Management: Redux, Zustand. Styling: Tailwind CSS, SASS. Testing: Jest, Cypress. Build Tools: Vite, Webpack.
- Full Stack Developer:** End-to-end integration across frontend, backend, and persistence tiers. Combined competency across React/Vue + Node.js/Python + relational/NoSQL datastores with Docker containerization.
- Mobile App Developer:** Cross-platform and native mobile toolchains: Flutter (Dart), React Native, Swift (iOS), Kotlin (Android). Local persistence (SQLite, Room, CoreData), push notifications, mobile app lifecycle.
- DevOps Engineer:** Infrastructure as Code (Terraform, Ansible), Container orchestration (Docker, Kubernetes), CI/CD pipelines (GitHub Actions, GitLab CI, Jenkins), Monitoring (Prometheus, Grafana), Cloud (AWS, Azure, GCP).

-
6. **Cybersecurity Analyst:** Security information & Event Management (SIEM), SOC monitoring, Vulnerability scanning, Penetration testing (Metasploit, Burp Suite), Cryptography, OWASP Top 10, Network packet analysis (Wireshark).
 7. **Data Scientist:** Exploratory data analysis, Statistical inference, Machine learning (Scikit-learn, XGBoost), Deep learning (PyTorch, TensorFlow), Feature engineering, Jupyter notebooks, Model evaluation metrics (ROC-AUC, F1).

APPENDIX B**COMPONENT 1 REST API SPECIFICATION & SCHEMAS**

The following JSON payload illustrates the output schema returned by the `/api/v1/cv/score` endpoint on Port 8001:

```
{
  "status": "success",
  "candidate_id": "CAND-2026-IT148-091",
  "predicted_role": "Backend Developer",
  "role_confidence": 0.942,
  "alternative_roles": [
    {"role": "Full Stack Developer", "probability": 0.038},
    {"role": "Software Engineer", "probability": 0.012},
    {"role": "DevOps Engineer", "probability": 0.005}
  ],
  "tri_pillar_scores": {
    "skill_alignment_score": 88.50,
    "experience_tenure_score": 92.00,
    "education_credential_score": 82.00,
    "composite_cv_fit_score": 88.25
  },
  "verified_tenure": {
    "total_professional_years": 4.50,
    "it_sector_years": 4.50,
    "target_role_relevant_years": 4.00,
    "overlapping_tenures_deduplicated": true
  },
  "extracted_entities": {
    "skills_count": 28,
    "workplace_evidence_skills": ["Python", "FastAPI", "PostgreSQL", "Docker", "Redis"],
    "isolated_mention_skills": ["Kubernetes", "AWS", "GraphQL"],
    "degree_level": "Bachelor of Science (Hons)",
    "degree_field": "Computer Science (Core Computing)",
    "certifications": ["AWS Certified Solutions Architect"]
  }
}
```

OpenAPI 3.0 Request Specification for `/api/v1/cv/screen``:

```
{
  "openapi": "3.0.3",
  "paths": {
    "/api/v1/cv/screen": {
      "post": {
        "summary": "Full Candidate Resume Screening & Tri-Pillar Fit Scoring",
        "requestBody": {
          "required": true,
          "content": {
            "multipart/form-data": {
              "schema": {
                "type": "object",
                "properties": {
                  "cv_file": {"type": "string", "format": "binary"},
                  "target_role": {"type": "string", "example": "Backend Developer"},
                  "job_description_text": {"type": "string", "default": ""},
                  "redact_pii": {"type": "boolean", "default": true}
                },
                "required": ["cv_file", "target_role"]
              }
            }
          }
        },
        "responses": {
          "200": {"description": "Successful screening evaluation with Tri-Pillar Scores and Roadmap"}
        }
      }
    }
  }
}
```

APPENDIX C**AUTOMATED 195-TEST SUITE EXECUTION REPORT**

Execution summary generated by the automated Pytest runner for Component 1:

```
===== test session starts =====
platform win32 -- Python 3.12.3, pytest-8.2.0, pluggy-1.5.0
rootdir: C:\Users\thari\Desktop\My Project\R26-IT-148
configfile: pytest.ini
collected 195 items

component1/tests/test_api.py ..... [ 7%]
component1/tests/test_candidate_intelligence_engine.py ..... [ 13%]
component1/tests/test_component1.py ..... [ 18%]
component1/tests/test_extractor.py ..... [ 35%]
component1/tests/test_international_education.py ..... [ 61%]
..... [ 77%]
component1/tests/test_job_specific_scores.py .... [ 79%]
component1/tests/test_matcher.py ..... [ 84%]
component1/tests/test_predictor.py ..... [ 89%]
component1/tests/test_scorer.py ..... [ 98%]
..... [100%]

===== 195 passed in 79.60s (00:01:19) =====
```